Human Powered Charging Bike Amazon Six Pager

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Introduction:

Cyclists around the world generate significant mechanical energy when riding, yet most of that energy goes unused. This project explores how all kinds of cycling can be transformed into a useful source of electrical energy. The Human-Powered Charging Bike (H.P.C.B.) is a portable, bike-mounted system that is designed to harvest human energy, We capture mechanical power from the rear wheel, regulate and store it through a custom electrical circuit, and display real-time ride data using an ESP32 microcontroller display and custom-made website. Our system includes a custom CAD-designed holder that organizes the battery pack and electronics, ensuring portability, safety, and ease of use. This custom 3D-printed housing also protects them from vibration and enables simple mounting and removal from the bike. Various forms of testing prove that the concept and product are fully integrated and functional. Testing also revealed inevitable voltage fluctuations at low riding speeds, that mechanical vibrations did somewhat affect stability, and that there were inconsistencies with the wireless connectivity. These challenges however introduced scopes to aim for and fix which led us to a more robust design targeting these weaknesses. While continuous website syncing remains partially implemented, the H.P.C.B. provides a reliable, user-friendly solution for riders who want a sustainable charging experience.

Key Points:

- Converts rotational energy → electrical energy using a converting motor
- Battery pack stores harvested energy
- ESP32 + screen provides data on the current voltage being generated
- ESP32 communicates with a website that displays voltage generated and has a built in sleep mode
- Custom 3D-printed holder mounts electronics securely to the bike

Goals:

The initial goal when creating this project was simple: to charge your phone while pedaling your bike. In engineering terms, that simply meant converting your mechanical input from pedaling into rotational energy, which is then transformed again into electrical power that is used to charge your phone. Taking into consideration that we had an entire electrical circuit to work with, we developed new goals that we wanted to accomplish:

- Improve Customer Experience: Ensure that the battery pack won't rattle and make a lot of noise when the bike is being ridden. Keep the system lightweight so it's easy to carry.

- Reliable Charge: Using sensors, control the voltage below a certain amount of volts to not fry the system.

- Increase Engagement: A screen to display charge data in real-time; A website to show fun facts about your ride such as watts generated.

- Conversation Starter: Make sure that the design is appealing to the eye; Want this product to be a proud purchase that is easy to recommend to a friend.

- Easy to use: Allow for the riders to attach, remove, charge, and carry the H.P.C.B. system effortlessly and intuitively.

- Safe Operation: Maintain a stable electrical and mechanical behavior across a wide range of environmental conditions, whether it be raining or a bumpy road.

The H.P.C.B. was designed so that a complete energy-harvesting experience was delivered to cyclists. These goals provided a shape and foundation for the system's architecture, including the circuitry, display interface, and physical design.

Tenets:

Our tenets allow us to keep the project aligned with our core engineering values: simplicity, safety, efficiency, and user-focused design. These principles guide decisions and keep the design workable and robust.

- Simplicity: Use as few components as possible to reduce weight, cost, and failure points. Make sure that the operation is extremely intuitive, i.e. all ages can use this product.

- Robustness: The design must withstand real-world cycling conditions (vibration, moisture, temperature, rain, debris).

- Efficiency: Maximize the conversion efficiency from mechanical to electrical energy, and reduce losses in the circuit; make sure no parts of circuit are drawing power when not needed.

- User-Centered Feedback: Real-time feedback is critical. Riders want to know how effective their pedaling is in terms of energy generation which needs to be in real-time.

- Scalability / Modularity: The system should be modular (power regulation, storage, display) so that future upgrades or replacements are easy.

- Safety: Electrical and mechanical systems must be reliably safe — no shorting, overheating, or danger to the rider. This includes voltage regulation, thermal protection, a secure enclosure, and protected wiring.

State of Business:

As of December 2025, the H.P.C.B. system is a fully functioning, production-ready product. The circuit, ESP32, and CAD holder all work individually and together. The mechanical, electrical, and software refinement which aimed to improve system performance ensure reliability and ease of operation.

Current System Status:

- Mechanical:
    - The rear-wheeled attached H.P.C.B. integrates cleanly onto a variety of bicycle frames and the wheel-to-wheel connecting rod is adjustable for different tire types as well.
    - Custom 3D-printed CAD enclosure is fully printed; securely fits battery + circuits + screen.
    - Rubber-damped additions minimize the vibrations and snuff out rattling.
    - Materials and geometry are designed for durability, weather resistance, and stability of the rider.

- Electrical:
    - High-efficiency buck converter and battery pack reliably regulate voltage at high speeds so as to not fry any electrical components. Even lower speeds will allow reliable charging as long as the bike is not completely stopped.
    - The diodes and protective circuity help to prevent reverse current which ensures safe operation.

    - The battery system can store excess energy and this will allow it to provide reliable charging even during coasting.

- Software / Data:
    - The ESP32 microcontroller measures voltage, current, power, and charge in real-time and sends this information to a website for user access.
    - The screen displays real-time data of the bike ride. The data contains information that the biker would want to know such as if the website connection is maintained and the amount of volts they are producing,
    - The wifi connection successfully links real-time ride data with the website. The user can use the website to alter settings that suit them such as how long after stopping the system should enter sleep mode.

- Testing Results:
    - Bench tests (bike is in control element and wheel is free spinning) show perfect wheel-to-wheel contact. Real-world testing meets expectations of slight vibration and voltage fluctuations but the system still works.
    - Waterproofing sealant successfully blocks out rain after controlled water pouring tests.

The current system of the H.P.C.B. serves all different types of cyclists in the form of a sustainable charging device. The product allows you to always have a battery pack in case you forgot to charge your phone in the morning or you are in an emergency situation. Its small footprint, durability, and rider analytics make it an appealing system for any bike user. The final product is quiet, reliable, simple to use, and operates seamlessly during normal riding.

Lessons Learned:

Early testing highlighted several engineering challenges related to voltage stability, mechanical vibration, thermal loading, and data reliability. These lessons directly shaped our next priorities and design adjustments. The development of the H.P.C.B. reinforced various engineering skills and principles. These lessons, skills, and priorities helped us directly shape the final product.

Key Lessons Learned:

- Voltage Output Fluctuates Widely: A large part of efficient charging depends on the bike rider and their speed. We used to have the voltage capped at 5V but realized that since we can safely harness more than this we made the cap around 20V. We also wanted to avoid the system from shutting off after stopping for too long or going too slowly so we have the system set to run even if generating only 0.2V. Mechanical vibrations can also affect

the output so more dampeners are needed.

- Thermal Rise Is Higher Than Expected: Buck converter tends to run very warm after a long ride. We may need a heat spreader or airflow design in future designs but since this might affect the waterproofing feature, a small fan integrated in might work. This also brings a few troubles as it is another feature that consumes power and would make the design bigger and heavier.

- ESP32 Power Consumption Is High: Adding sleep mode was a great way to drastically decrease power consumption when the user was not riding the machine. Because the power consumption is still non-zero however, adding a deep-sleep mode might be a way to further improve the system. Because the system automatically turns back on after biking for a few seconds, the sleep mode is the best solution for the ESP32's consumption.

- WiFi Connectivity Drops During Movement: Due to the nature of the ESP32 which requires wifi, a moving system which naturally has varying distance to the wifi source, will fluctuate in connectivity. As there is no solution for a router which doesn't move, any users that will want to use the live-feed feature will need to use their phone's hotspot. Because the charging still works without connection, this problem is relatively minor however.

- User Interface Needs Simplification And Additions: The current website which displays data and personalization is too cluttered. We need larger fonts, fewer metrics on the main screen, and also intuitive icons that will help the user find what they are looking for in an appealing fashion. We also need to add additional key performance metrics such as miles ridden which most bikers would care more about.

Strategic Priorities:

To move toward a polished final product, our next steps emphasize stability, durability, firmware optimization, and user experience. Field testing and data analysis will confirm our design choices and guide final improvements. Each category is a field we could aim to improve.

Mechanical Improvements:

- Edit CAD holder; make the system lighter with similar mechanical properties through either different material choice or geometry of holder.
- Add more rubber dampers for additional vibration absorption.

- Improve wheel-to-wheel contact alignment and wheel material. Current wheel material has a short usage lifespan so material could be exchanged to a gripper, sturdier material.

Electrical Refinements:

- Test more efficient buck converters, current converters being used are cheapest ones.
- Add some sort of mechanism for thermal management. Heat sinks would be the best option as they don't consume much energy. Keeping weight the same would be ideal but most important features would be electrical circuit not overheating.

Firmware & Interface Upgrades:

- Implement ESP32 deep-sleep modes as current sleep mode still consumes some power.
- Add local storage (SPIFFS or SD card) for ride logs. This way the user could go onto the website and see their past rides for improvement.
- Redesign simple, readable ride dashboard. Display more stats that would pertain to a user who does not care about the technical stuff.

Validation & Long-Term Durability:

- Stress/field testing: Vibration table, water spray, shock drop tests, dusty environment.
- Record rider feedback: We should ask multiple people, preferably bikers, to ride with the H.P.C.B. on and report on their comfortability and thoughts on the design.
- Improve strain relief for all wiring harnesses: The current design mainly uses glue to hold down wires and act as a vibration absorbent but better harnessing could help with durability.

Additional add on features or optional modifications:

- Phone holder: This would be a phone holder on the handlebars which would connect to the battery pack so the user could see their phone as they ride. This might be helpful for delivery riders and this product could also easily be made waterproof.
- Second wheel: We would make a design that would have two points of contact on the backwheel instead of one. The design would simply have a second motor. While the weight would be doubled, so would power generated; if a user wants more power and doesn't care about weight this would be a good option for them.
- Various models depending on weather: Because this product can be used globally, it would make sense to make models based on location. In Seattle where it rains often and is colder, waterproofing and worrying about overheating is a smaller concern. In a hotter country, air vents could be made and the device could be made lighter without waterproofing features.

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Appendix A: Market Mission and Systems Requirements

Market Mission:

The Human Powered Charging Bike (H.P.C.B.) aims to be a useful amenity for regular bikers and commuters. Currently, energy is generated and used by cyclists, but it is almost never harvested beyond movement of the bicycle. Excess energy is essentially wasted, especially in scenarios where the rider does not need to pedal (downhill, momentum, etc.). The goal of the H.P.C.B. is to convert a portion of energy from cycling into useful electrical energy for charging electronic devices in an unobtrusive way. The primary requirements for this mission are for the H.P.C.B. to be light, provide little-to-no resistance, and be almost universally mountable onto any bicycle. Specific physical requirements are outlined in Appendix G, in the "Revised (As Offered)" column of Table G1.

System Requirements:

The specific system requirements are as follows:

- The H.P.C.B. shall produce a useful amount of Wattage when mounted on a bike going 10 mph. Useful is defined as 5 Volts at 0.5 Amps for the purposes of this project.
- The H.P.C.B. shall be able to convert the Voltage it generates into a battery percentage, and report it back to the user.

Both of these specific system requirements were met in the prototype, as shown in Figure A1 and A2. The battery that was used for this product does not begin to charge until it can draw 0.5 Amps. Therefore, in figure A1, the only significant value is Voltage, as it allows for a simple Wattage conversion. Additionally, the calculations shown in Figure A2 use 0.5 Amps as a constant measured current.

Human Power Charging Bike Power Output

Measured values:

$Output\ Voltage = V = 5\ V$ $\qquad$ $Rotations\ for\ 5\ V = \ rpm_{5V} = 1200\ rpm$

$Bike\ Wheel\ Diameter = d_{bike} = 27.5\ in$ $\qquad$ $Friction\ Roller\ Diameter = d_{roller} = 3.6\ cm$

Finding rotations per second ($\frac{rotations}{s}$) as a function of bike speed (v):

$$\frac{rotations}{s} = \frac{v}{d_{bike} \times \pi} = \frac{v}{27.5\ in \times \frac{m}{39.37\ in} \times 3.142} = \frac{v}{2.19\ m}$$

Finding rotations per second of the friction roller ($\frac{rotations}{s}_{roller}$) in terms of rotations per second ($\frac{rotations}{s}$):

$$\frac{rotations}{s}_{small} = \frac{d_{bike}}{d_{roller}} \times \frac{rotations}{s} = \frac{27.5\ in\ \times \frac{m}{39.27\ in}}{3.6\ cm \times \frac{m}{100\ cm}} \times \frac{v}{2.19\ m} = 8.88 \times v\ \frac{rotations}{s}$$

Finding output voltage (V) in terms of rotations per second of the friction roller ($\frac{rotations}{s}_{roller}$) from the measured values (assuming voltage is linearly proportional):

$V = \frac{rotations}{s}_{roller} \times k$, where k is the proportionality is constant between voltage and friction roller rotations.

Plugging in known values allows k to be found.

$$k = \frac{V}{rpm_{5V}} = \frac{5\ V}{1200\ rpm \times \frac{rpm}{60\ \frac{rotations}{s}}} = .25\ \frac{V}{\frac{rotations}{s}}$$

Combining equations to solve for voltage (V) as a function of bike speed (v):

$$V = k\ \times \frac{rotations}{s}_{roller} = \frac{.25\ V}{\frac{rotations}{s}} \times 8.88 \times v\ \frac{rotations}{s} = 2.22 \times\ v$$

This gives the final relationship between output voltage in volts (V) and bike speed in meters per second (v):

$$V = 2.22 \times v$$

<u>Assumption:</u> There is no slippage between the friction roller and the bike wheel.

Example calculation: Voltage generated at a bike speed of 10 mph:

$$10\ mph \times \frac{.447\ \frac{m}{s}}{mph} \times 2.22 = 9.92\ V$$

This value lines up with values obtained in our testing with the device mounted on a 27.5 in wheel bike. To adjust the relationship for different sized bikes, d_{bike} would be change, changing the ratio between the two wheels. However, 27.5 in is the most common bike wheel size for adult bikes and is where the device would most likely be used.

<u>Figure A1: Voltage Output as a Function of Bike Speed</u>

Battery Charge Calculations

Measured Values

$Charging\ Current: I_{charge} = .5\ A$

$ESP32\ Current\ Draw: I_{ESP32} = 200\ mA$

Manufacturer Values

$ESP32\ Voltage: V_{ESP32} = 3.3\ V$

$Nominal\ Battery\ Cell\ Volatge: V_{cell} = 3.7\ V$

$Battery\ Capacity\ mAh = C_{mAh} = 10000\ mAh$

Finding Battery Capacity (C) in Watt-hours:

$$C = V_{cell} \times C_{mAh} = 3.7\ V \times 10000\ mAh \times \frac{1\ Ah}{1000\ mAh} = 37\ Wh$$

Finding Power generated in Watts, accounting for energy usage by the ESP32:

$$P = (I_{charge} \times V) - (I_{ESP32} \times V_{ESP32}) = .5 \times V - (.5\ A \times 3.3\ V) = .5\ A \times V - 1.65\ W$$

Using time spent charging in hours (t) recorded by the ESP32 and voltage generated in Volts (V), measured by the sensor, to calculate percentage of the battery capacity generated while the dynamo was in use:

$$Energy\ Gained = P \times t = (.5\ A \times V - 1.65)\ W \times t$$

$$Percent\ gained = \frac{Energy\ Gained}{C} \times 100 = \frac{(.5\ A \times V - 1.65)\ W \times t}{37\ Wh} \times 100$$

This is a very simplified approximation, since battery chemistry is very complicated, and measurements would be hard to get accurate beyond a simple approximation without dismantling the battery (which needs to remain easily removable per project requirements). For the recreational purposes of charging a battery while riding your bike, this approximation is adequate.

Example Calculation: Bike ride at 10 mph for 2 hours:

$V = 2.22 \times v = 2.22 \times 10\ mph \times \frac{.447\frac{m}{s}}{mph} = 9.92\ V$ (derived voltage to bike speed relationship)

$$Percent\ gained = \frac{(.5\ A \times V - 1.65)\ W \times t}{37\ Wh} \times 100 = \frac{(.5\ A \times 9.92\ V - 1.65)W \times 2\ h}{37\ Wh} \times 100 = 17.9\%$$

If the bike was ridden at 10 mph (relatively low biking speed) for 2 hours, 17.9% of the battery capacity would be gained.

This calculation is being performed by the ESP32 to report a percentage gained value.

Figure A2: Battery Charge Calculations

The values determined in Figure A1 and A2 demonstrate how both of the specific system requirements are met, with the power output requirement being met at almost double the rate of the original system goal. Each of the results from the calculations were cross referenced with testing data, and found to be accurate.

Appendix B: Business Model

Product: The product that we sell is called the H.P.C.B., short for human powered charging bike, and it is a human-powered battery charger designed to fit onto any bike and keep a battery pack charged for convenience or in case of emergencies. The appeal of the H.P.C.B. is that it is extremely portable and lightweight, easy to use, and charges your battery pack whenever you are on the go very efficiently. Our product would be cheap to buy costing only around 80$ and can be used the second you unpack it from the box.

Workers: Due to the simplicity of our design, the H.P.C.B. would only need assemblers if we were to scale up the factory. If we wanted to make all of the electrical components in house then we would also need operators and technicians. The assembler's main jobs would be to connect the various electrical components to the casing shell and tighten any screws/bolts that are included in the design. The assembly line would then pass it on to the shipping department for final deliveries to your door or to an in store location. None of the labor would be physically intensive and it is possible that a large amount of it could be automated.

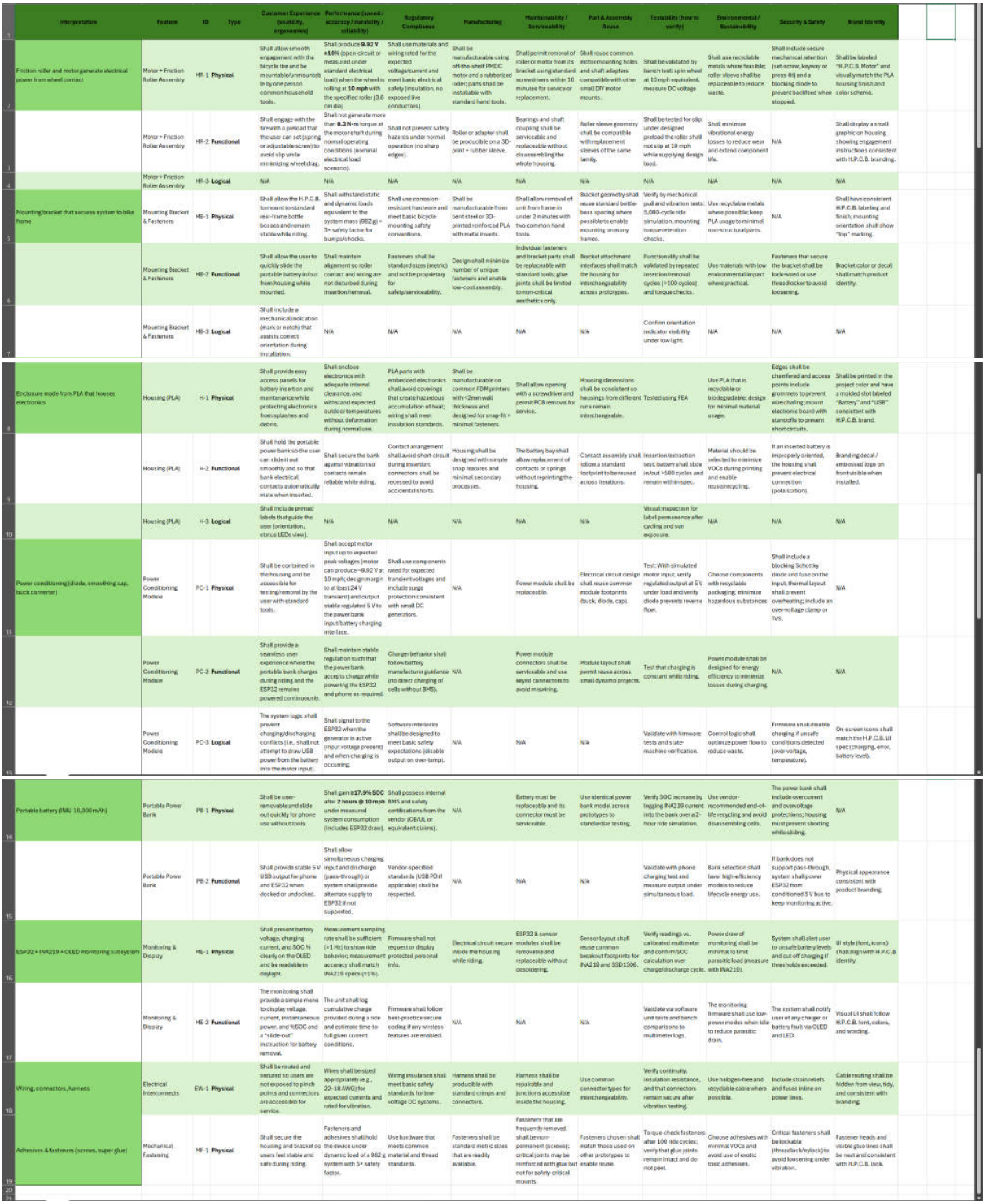
Tools & Equipment: The H.P.C.B. is so cheap because the entirety of the case of other various connector parts can be 3D printed and are made with filament. The main tools and equipment that would be needed would simply be rows of 3D printers and PLA filament. There might be other various tools needed for the electrical components and fasteners such as a soldering iron, screws, bolts, allen wrenches, etc, but much of the work is done and performed by 3D printers. If the electrical components were wanting to be made in shop or if the case would want to be upgraded in material despite cost raises, then an injection molding machine would also be needed. For the current product however, not much equipment past a 3D printer and soldering iron would be needed.

Production flow: Production would start with a simple shipping of everything needed to create the H.P.C.B. The material list would start with PLA filament, including all the various electrical components (diodes, wires, display screen, ESP32, etc.), and finally the various fasteners/sealants. The filament would be loaded onto normal sized 3D printers and being a decently long print that would include the case, the electrical component holder, the battery pack holder, and the motor to bike wheel connecting rod. After the 3D print is completed, workers will be tasked with soldering the electrical circuit together and then connecting the completed electrical circuit to the electrical component sleeve in the case using either glue or wire holders. Every wire and pin connection will be soldered and wrapped in rubber. The screen will be glued into place and the battery pack will be slid in while connecting it to the ESP32. All the fasteners will be screwed in firmly connecting all 3D parts together. Finally sealant and rubber gaskets will be added to make the design waterproof and vibration absorbent. The finished product will then be shipped out to the customer, ready to be attached to the bike with already given fasteners and directions of use.

Market context: Who the product is for, approximate production scale, demand level, etc.

The H.P.C.B. has a very clear target market: bikers. Due to the nature of the mechanism, it only works on and is effective on bikes. While it might be possible to add it to other modes of transport, this product is tailored for bikers who often find themselves running low on phone battery when on low rides. The scale level of our production systems is estimated to be around 10 completed products a day which would be accomplishable by around 5 machines printing back to back and only a few specialized workers. The demand level for a product like this should be high as many bikers desire the ease of mind knowing their phone won’t die.

Appendix C: Model Based Systems Engineering



Interpretation	Feature	ID	Type	Customer Experience (usability, ergonomics)	Performance (speed / accuracy / durability / reliability)	Regulatory Compliance	Manufacturing	Maintainability / Serviceability	Part & Assembly Reuse	Testability (how to verify)	Environmental / Sustainability	Security & Safety	Brand Identity
Friction roller and motor generate electrical power from wheel contact	Motor + Friction Roller Assembly	MR-1	Physical	Shall allow smooth engagement with the bicycle tire and be mountable/unmountable by one person common household tools.	Shall produce **9.92 V ±10%** (open-circuit or measured under standard electrical load) when the wheel is rolling at **10 mph** with the specified roller (3.8 cm dia).	Shall use materials and wiring rated for the expected voltage/current and meet basic electrical safety (insulation, no exposed live conductors).	Shall be manufacturable using off-the-shelf PMDC motor and a rubberized roller; parts shall be installable with standard hand tools.	Shall permit removal of roller or motor from its bracket using standard screwdrivers within 10 minutes for service or replacement.	Shall reuse common motor mounting holes and shaft adapters compatible with other small DIY motor mounts.	Shall be validated by bench test: spin wheel at 10 mph equivalent, measure DC voltage	Shall use recyclable metals where feasible; roller sleeve shall be replaceable to reduce waste.	Shall include secure mechanical retention (set-screw, keyway or press-fit) and a blocking diode to prevent backfeed when stopped.	Shall be labeled "H.P.C.B. Motor" and visually match the PLA housing finish and color scheme.
	Motor + Friction Roller Assembly	MR-2	Functional	Shall engage with the tire with a preload that the user can set (spring or adjustable screw) to avoid slip while minimizing wheel drag.	Shall not generate more than **0.3 N-m** torque at the motor shaft during nominal operating conditions (nominal electrical load scenario).	Shall not present safety hazards under normal operation (no sharp edges).	Roller or adapter shall be producible on a 3D-print + rubber sleeve.	Bearings and shaft coupling shall be serviceable and replaceable without disassembling the whole housing.	Roller sleeve geometry shall be compatible with replacement sleeves of the same family.	Shall be tested for slip: under designed preload the roller shall not slip at 10 mph while supplying design load.	Shall minimize vibrational energy losses to reduce wear and extend component life.	N/A	Shall display a small graphic on housing showing engagement instructions consistent with H.P.C.B. branding.
	Motor + Friction Roller Assembly	MR-3	Logical	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mounting bracket that secures system to bike frame	Mounting Bracket & Fasteners	MB-1	Physical	Shall allow the H.P.C.B. to mount to standard rear-frame bottle bosses and remain stable while riding.	Shall withstand static and dynamic loads equivalent to the system mass (982 g) × 3× safety factor for bumps/shocks.	Shall use corrosion-resistant hardware and meet basic bicycle mounting safety conventions.	Shall be manufacturable from bent steel or 3D-printed reinforced PLA with metal inserts.	Shall allow removal of unit from frame in under 2 minutes with two common hand tools.	Bracket geometry shall reuse standard bottle-boss spacing where possible to enable mounting on many frames.	Verify by mechanical pull and vibration tests: 5,000-cycle ride simulation, mounting torque retention checks.	Use recyclable metals where possible; keep PLA usage to minimal non-structural parts.	N/A	Shall have consistent H.P.C.B. labeling and finish; mounting orientation shall show "top" marking.
	Mounting Bracket & Fasteners	MB-2	Functional	Shall allow the user to quickly slide the portable battery in/out from housing while mounted.	Shall maintain alignment so roller contact and wiring are not disturbed during insertion/removal.	Fasteners shall be standard sizes (metric) and not be proprietary for safety/serviceability.	Design shall minimize number of unique fasteners and enable low-cost assembly.	Individual fasteners and bracket parts shall be replaceable with standard tools; glue joints shall be limited to non-critical aesthetics only.	Bracket attachment interfaces shall match the housing for interchangeability across prototypes.	Functionality shall be validated by repeated insertion/removal cycles (≥100 cycles) and torque checks.	Use materials with low environmental impact where practical.	Fasteners that secure the bracket shall be lock-wired or use threadlocker to avoid loosening.	Bracket color or decal shall match product identity.
	Mounting Bracket & Fasteners	MB-3	Logical	Shall include a mechanical indication (mark or notch) that assists correct orientation during installation.	N/A	N/A	N/A	N/A	N/A	Confirm orientation indicator visibility under low light.	N/A	N/A	N/A
Enclosure made from PLA that houses electronics	Housing (PLA)	H-1	Physical	Shall provide easy access panels for battery insertion and maintenance while protecting electronics from splashes and debris.	Shall enclose electronics with adequate internal clearance, and withstand expected outdoor temperatures without deformation during normal use.	PLA parts with embedded electronics shall avoid coverings that create hazardous accumulation of heat; wiring shall meet insulation standards.	Shall be manufactureable on common FDM printers with ~12mm wall thickness and designed for snap-fit + minimal fasteners.	Shall allow opening with a screwdriver and permit PCB removal for service.	Housing dimensions shall be consistent so housings from different runs remain interchangeable.	Tested using FEA	Use PLA that is recyclable or biodegradable; design for minimal material usage.	Edges shall be chamfered and access points include grommets to prevent wire chafing; mount electronic board with standoffs to prevent short circuits.	Shall be printed in the project color and have a molded slot labeled "Battery" and "USB" consistent with H.P.C.B. brand.
	Housing (PLA)	H-2	Functional	Shall hold the portable power bank so the user can slide it out smoothly and so that bank electrical contacts automatically mate when inserted.	Shall secure the bank against vibration so contacts remain reliable while riding.	Contact arrangement shall avoid short-circuit during insertion; connectors shall be recessed to avoid accidental shorts.	Housing shall be designed with simple snap features and minimal secondary processes.	The battery bay shall allow replacement of contacts or springs without reprinting the housing.	Contact assembly shall follow a standard footprint to be reused across iterations.	Insertion/extraction test: battery shall slide in/out >500 cycles and remain within spec.	Material should be selected to minimize VOCs during printing and enable reuse/recycling.	If an inserted battery is improperly oriented, the housing shall prevent electrical connection (polarization).	Branding decal / embossed logo on front visible when installed.
	Housing (PLA)	H-3	Logical	Shall include printed labels that guide the user (orientation, status LEDs view).	N/A	N/A	N/A	N/A	N/A	Visual inspection for label permanence after cycling and sun exposure.	N/A	N/A	N/A
Power conditioning (diode, smoothing cap, buck converter)	Power Conditioning Module	PC-1	Physical	Shall be contained in the housing and be accessible for testing/removal by the user with standard tools.	Shall accept motor input up to expected peak voltages (motor can produce ~9.92 V at 10 mph; design margin to at least 24 V transient) and output stable regulated 5 V to the power bank input/battery charging interface.	Shall use components rated for expected transient voltages and include surge protection consistent with small DC generators.	N/A	Power module shall be replaceable.	Electrical circuit design shall reuse common module footprints (buck, diode, cap).	Test: With simulated motor input, verify regulated output at 5 V under load and verify diode prevents reverse flow.	Choose components with recyclable packaging; minimize hazardous substances.	Shall include a blocking Schottky diode and fuse on the input; thermal layout shall prevent overheating; include an over-voltage clamp or TVS.	N/A
	Power Conditioning Module	PC-2	Functional	Shall provide a seamless user experience where the portable bank charges during riding and the ESP32 remains powered continuously.	Shall maintain stable regulation such that the power bank accepts charge while powering the ESP32 and phone as required.	Charger behavior shall follow battery manufacturer guidance (no direct charging of cells without BMS).	N/A	Power module connectors shall be serviceable and use keyed connectors to avoid miswiring.	Module layout shall permit reuse across small dynamo projects.	Test that charging is constant while riding.	Power module shall be designed for energy efficiency to minimize losses during charging.	N/A	N/A
	Power Conditioning Module	PC-3	Logical	The system logic shall prevent charging/discharging conflicts (i.e., shall not attempt to draw USB power from the battery into the motor input).	Shall signal to the ESP32 when the generator is active (input voltage present) and when charging is occurring.	Software interlocks shall be designed to meet basic safety expectations (disable output on over-temp).	N/A	N/A	N/A	Validate with firmware tests and state-machine verification.	Control logic shall optimize power flow to reduce waste.	Firmware shall disable charging if unsafe conditions detected (over-voltage, temperature).	On-screen icons shall match the H.P.C.B. UI spec (charging, error, battery level).
Portable battery (INIU 10,000 mAh)	Portable Power Bank	PB-1	Physical	Shall be user-removable and slide out quickly for phone use without tools.	Shall gain ≥**17.0% SOC** after **2 hours @ 10 mph** under measured system consumption (includes ESP32 draw).	Shall possess internal BMS and safety certifications from the vendor (CE/UL or equivalent claims).	N/A	Battery must be replaceable and its connector must be serviceable.	Use identical power bank model across prototypes to standardize testing.	Verify SOC increase by logging INA219 current into the bank over a 2-hour ride simulation.	Use vendor-recommended end-of-life recycling and avoid disassembling cells.	The power bank shall include overcurrent and overvoltage protections; housing must prevent shorting while sliding.	N/A
	Portable Power Bank	PB-2	Functional	Shall provide stable 5 V USB output for phone and ESP32 when docked or undocked.	Shall allow simultaneous charging input and discharge (pass-through) or system shall provide alternate supply to ESP32 if not supported.	Vendor-specified standards (USB PD if applicable) shall be respected.	N/A	N/A	N/A	Validate with phone charging test and measure output under simultaneous load.	Bank selection shall favor high-efficiency models to reduce lifecycle energy use.	If bank does not support pass-through, system shall power ESP32 from conditioned 5 V bus to keep monitoring active.	Physical appearance consistent with product branding.
ESP32 + INA219 + OLED monitoring subsystem	Monitoring & Display	ME-1	Physical	Shall present battery voltage, charging current, and SOC % clearly on the OLED and be readable in daylight.	Measurement sampling rate shall be sufficient (≥1 Hz) to show ride behavior; measurement accuracy shall match INA219 specs (±1%).	Firmware shall not request or display protected personal info.	Electrical circuit secure inside the housing while riding.	ESP32 & sensor modules shall be removable and replaceable without desoldering.	Sensor layout shall reuse common breakout footprints for INA219 and SSD1306.	Verify readings vs. calibrated multimeter and confirm SOC calculation over charge/discharge cycle.	Power draw of monitoring shall be minimal to limit parasitic load (measure with INA219).	System shall alert user to unsafe battery levels and cut off charging if thresholds exceeded.	UI style (font, icons) shall align with H.P.C.B. identity.
	Monitoring & Display	ME-2	Functional	The monitoring shall provide a simple menu to display voltage, current, instantaneous power, and %SOC and a "slide-out" instruction for battery removal.	The unit shall log cumulative charge provided during a ride and estimate time-to-full given current conditions.	Firmware shall follow best-practice secure coding if any wireless features are enabled.	N/A	N/A	N/A	Validate via software unit tests and bench comparisons to multimeter logs.	The monitoring firmware shall use low-power modes when idle to reduce parasitic drain.	The system shall notify user of any charger or battery fault via OLED and LED.	Visual UI shall follow H.P.C.B. font, colors, and wording.
Wiring, connectors, harness	Electrical Interconnects	EW-1	Physical	Shall be routed and secured so users are not exposed to pinch points and connectors are accessible for service.	Wires shall be sized appropriately (e.g., 22–18 AWG) for expected currents and rated for vibration.	Wiring insulation shall meet basic safety standards for low-voltage DC systems.	Harness shall be producible with standard crimps and connectors.	Harness shall be repairable and junctions accessible inside the housing.	Use common connector types for interchangeability.	Verify continuity, insulation resistance, and that connectors remain secure after vibration testing.	Use halogen-free and recyclable cable where possible.	Include strain reliefs and fuses inline on power lines.	Cable routing shall be hidden from view, tidy, and consistent with branding.
Adhesives & fasteners (screws, super glue)	Mechanical Fastening	MF-1	Physical	Shall secure the housing and bracket so users feel stable and safe during riding.	Fasteners and adhesives shall hold the device under dynamic load of a 982 g system with 5× safety factor.	Use hardware that meets common material and thread standards.	Fasteners shall be standard metric sizes that are readily available.	Fasteners that are frequently removed shall be non-permanent (screws); critical joints may be reinforced with glue but not for safety-critical mounts.	Fasteners chosen shall match those used on other prototypes to enable reuse.	Torque-check fasteners after 100 ride cycles; verify that glue joints remain intact and do not peel.	Choose adhesives with minimal VOCs and avoid use of exotic toxic adhesives.	Critical fasteners shall be lockable (threadlock/nylock) to avoid loosening under vibration.	Fastener heads and visible glue lines shall be neat and consistent with H.P.C.B. look.

Figure C1: H.P.C.B. Required, Logical, Funcitonal

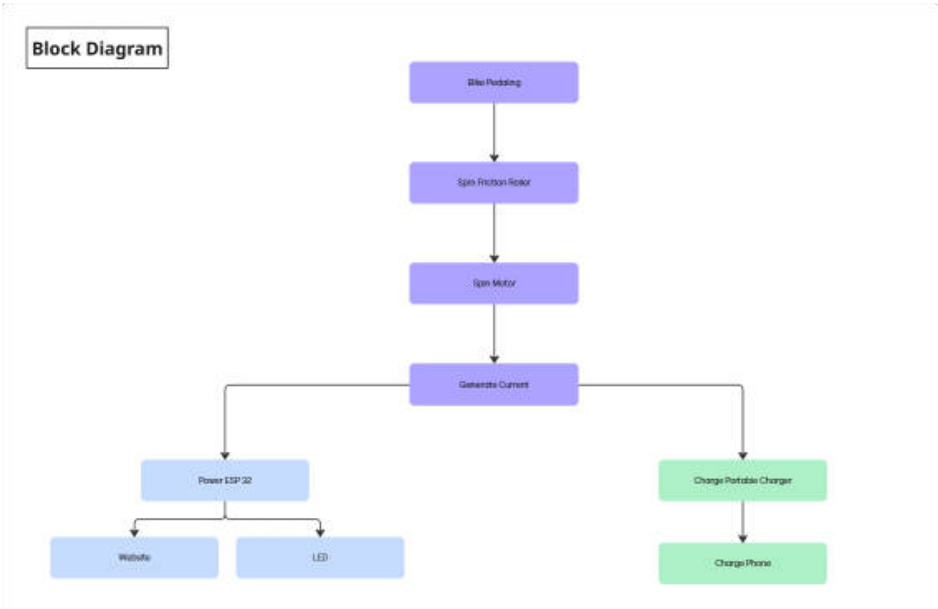


Figure C2: H.P.C.B. Block Diagram

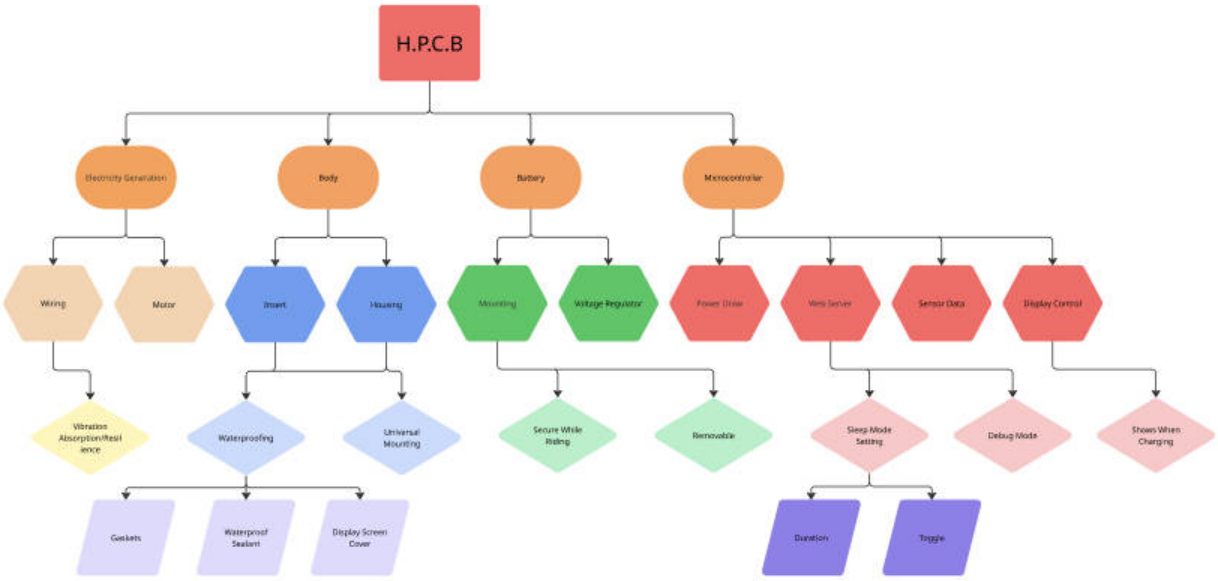


Figure C3: H.P.C.B. System Diagram

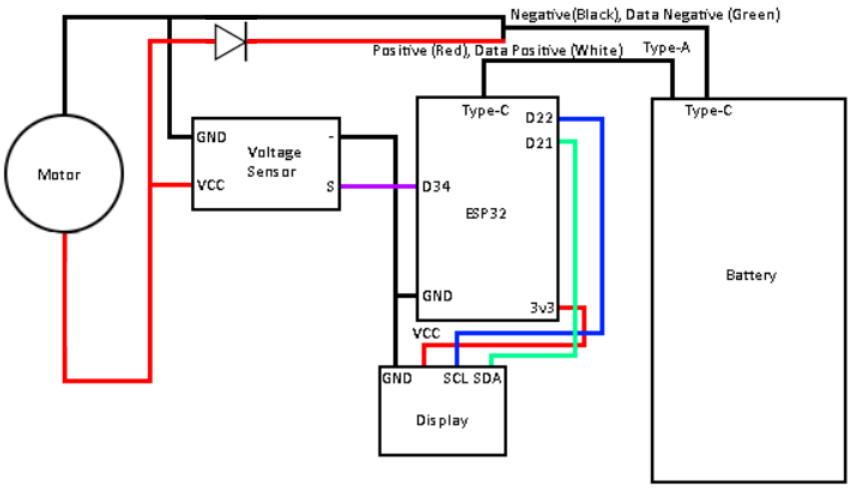


Figure C4: H.P.C.B. Electrical Diagram

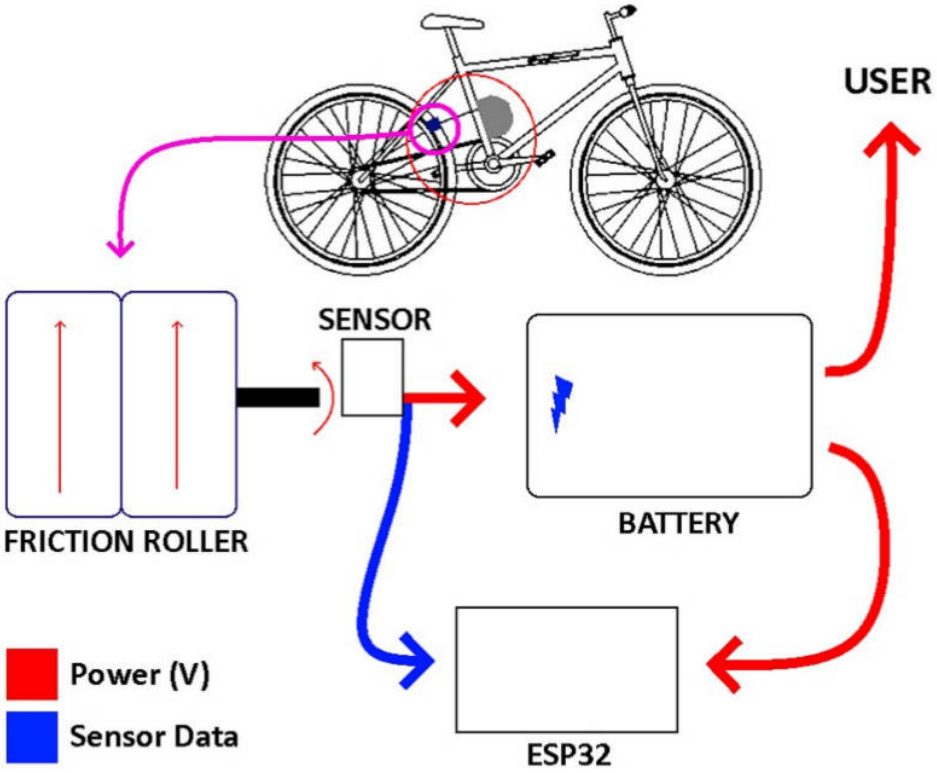


Figure C5: H.P.C.B. Flow Chart

Appendix D: Model Based Definition

Technical Computer Aided Design (CAD) Drawing:

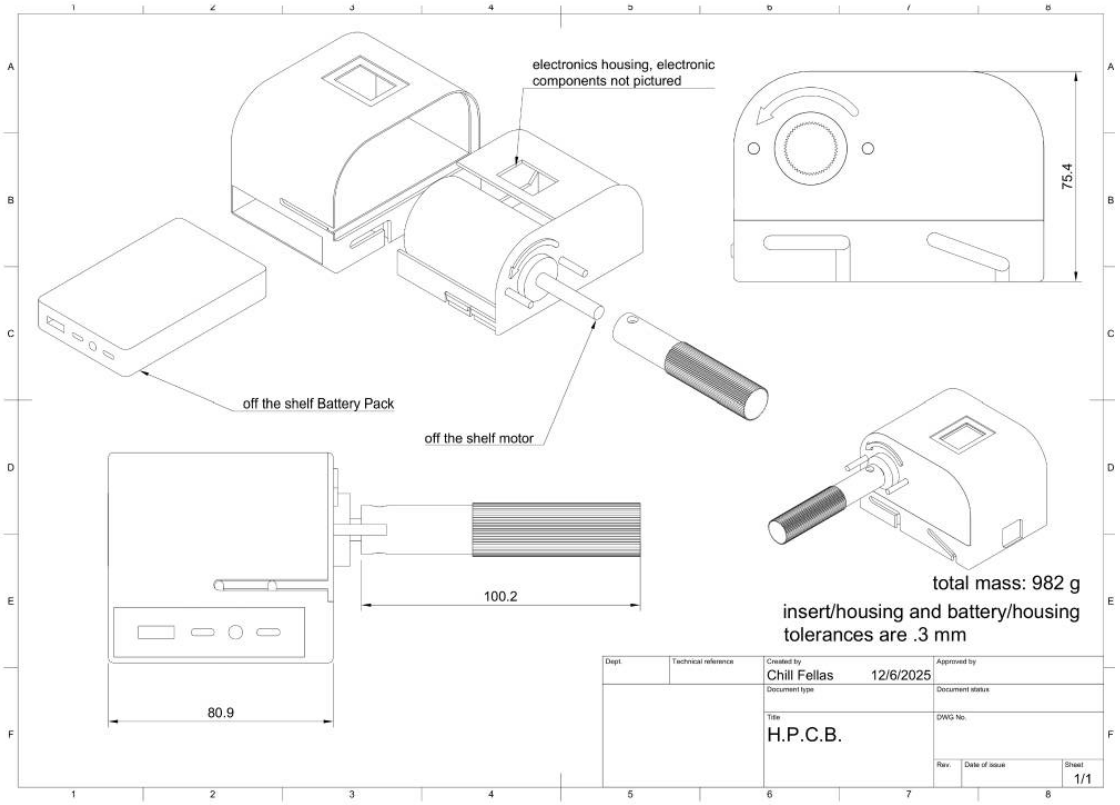


Figure D1: H.P.C.B. Technical Assembly Drawing

GD&T CAD Drawing:

Dimensioned drawings of each of the components are shown in Figures D2, D3, and D4. Since these components are each to be 3D printed or later moulded, the design is made for an FDM 3D printer rather than a machinist.

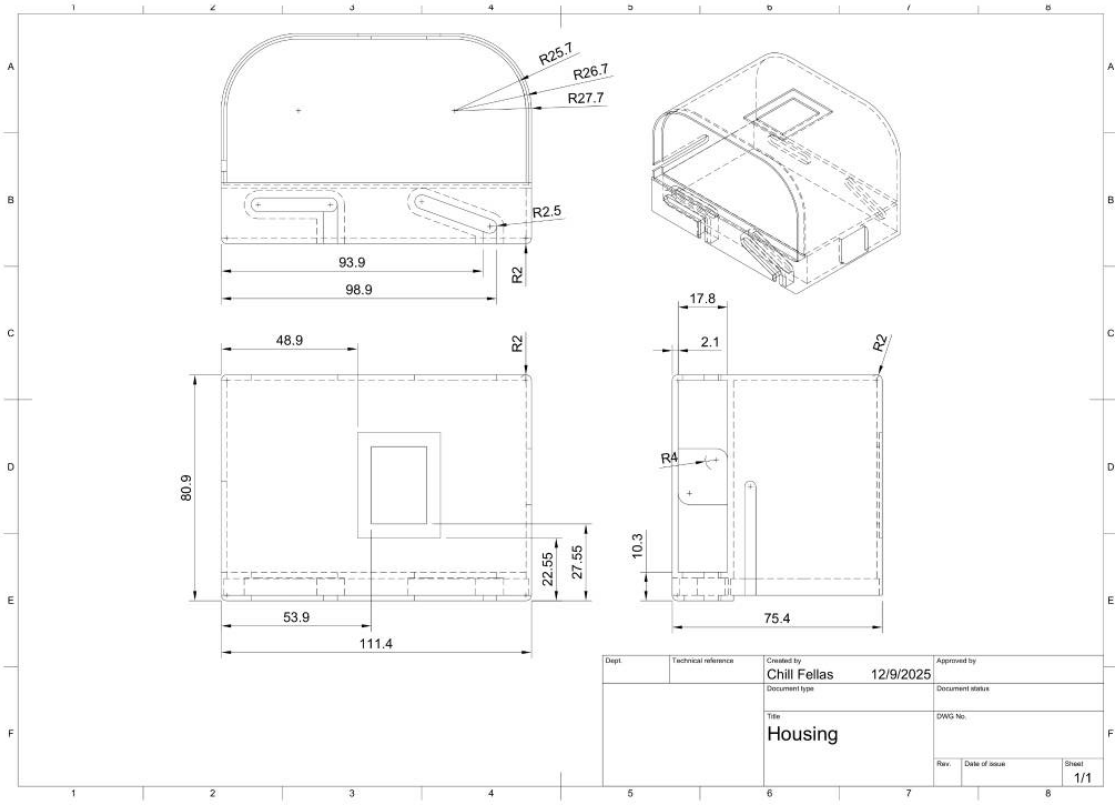


Figure D2: Housing Drawing

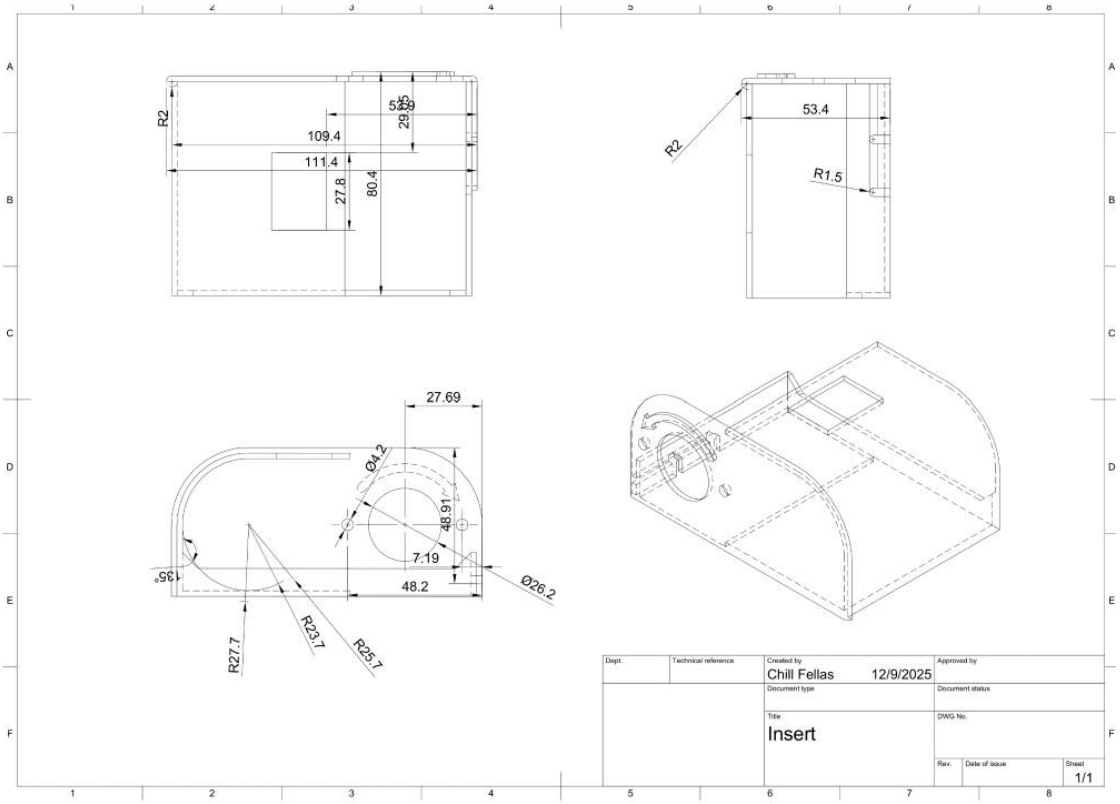


Figure D3: Insert Drawing

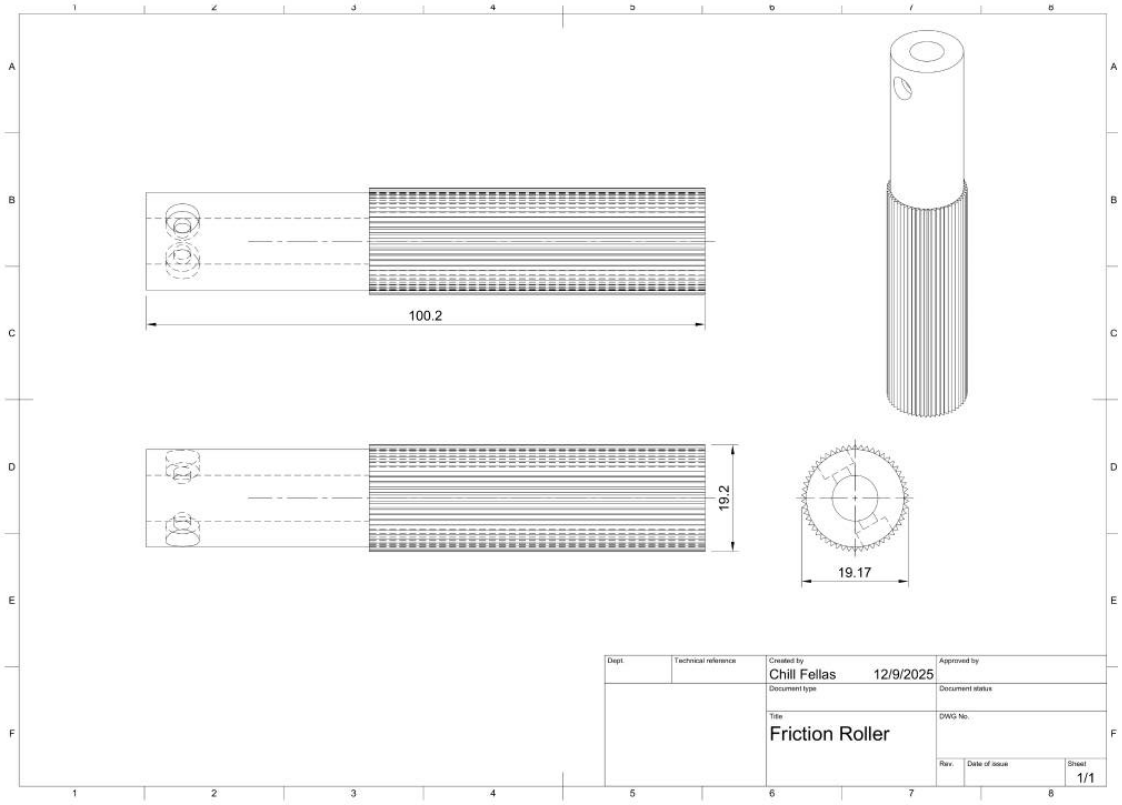
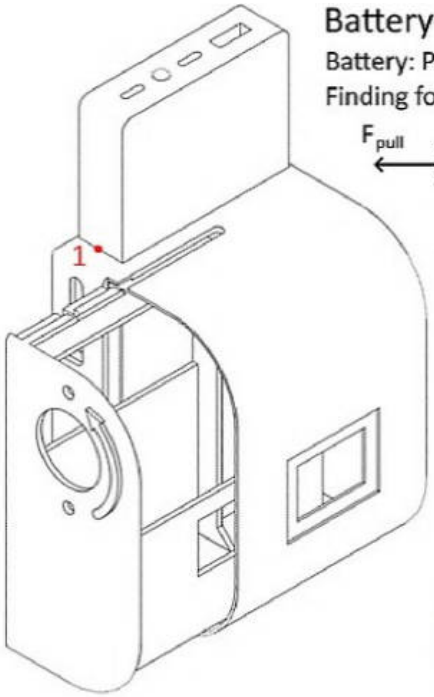


Figure D4: Friction Roller Drawing

Free Body Diagrams:

Free body diagrams were drawn to analyze the forces required to separate various components at their joints. The H.P.C.B. device has two connections that are dependent on friction. These are the connection of the battery to the housing, and the connection of the insert to the housing. The purpose of the friction connection in the case of the battery and housing is so that the battery is easily removable, and can therefore be used outside of the H.P.C.B. device. The purpose of the friction hold in the case of the insert and housing is so that the housing can be removed without the need of any tools, and the device can thus be easily repaired. This connection is also waterproofed to protect the electronics, which is done via the use of a gasket. Neither of these two connections should come undone under common use conditions. The free body diagrams for

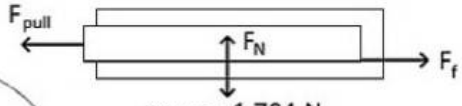
each of these connections are in Figure D5. In these diagrams, the mass of the friction roller is not included, as it is not a significant part of the force acting on either component.



Battery-Housing Connection

Battery: PC/ABS, Housing: PLA => $\mu_s = .4$, m = 182 g = .182 kg

Finding force of friction:



m x g = 1.784 N

$\sum F_y = 0 \Rightarrow F_N = 1.784$ N $\quad \sum F_x = 0 \Rightarrow F_{pull} = F_f \times \mu_s = .7136$ N

$F_f = F_N \times \mu_s = .7136$ N $= F_{pull}$

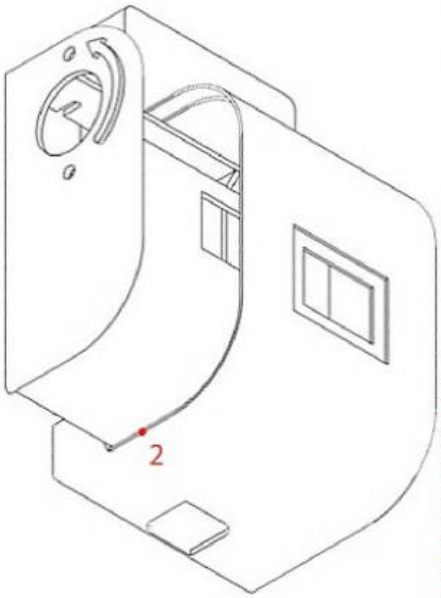
Finding force to pull battery out:



$F_N = 2.498$ N $= F_{pull\ vertical}$

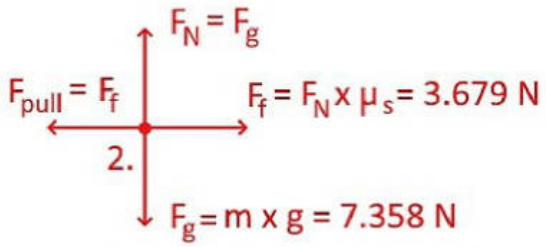
$F_g + F_f = 2.498$ N

To pull the battery out, it would take .7136 N in the horizontal configuration. When mounted on the bike, it is in the vertical configuration, where it would take about 2.5 N to remove the battery.



Insert-Housing Connection

Insert, PLA, Housing: PLA => $\mu_s = .5$, m = 750 g = .750 kg



The force required to pull the insert out of the housing when the dynamo is mounted on the bike (vertical position, horizontal pulling) is about 3.7 N. This number does not account for the rubber gasket on the housing, which has a static friction coefficient of about 1.0 with the PLA, and would bring the force required to pull the insert out to about 7.4 N. It is not included due to the potential of the adhesive failing at a force lower than 7.4 N.

Assumption: The battery/insert contact the housing only on the face facing the ground due to a clearance fit between parts (+ .3 mm).

Assumption: There is no bending due to stresses in the PLA components.

Figure D5: Free Body Diagrams of Friction Connections

Finite Element Models:

Finite element models were created using ANSYS. For the forces, we calculated the weight and the torque caused from the friction roller contacting the back wheel of the bike. The calculated torque only reached 0.3 Nm under the most extreme conditions. This is expected, as the design is made to be unobtrusive to the biker, and therefore should not add significant torque. The final mass of the H.P.C.B. device was 0.982kg. To simplify analysis, we rounded up this mass, multiplied by gravity, to get a 10 N of force as a result of the device's mass. To leave room for edge case scenarios, we used a FOS of a little over 3. With this, we applied a total of 1 Nm of torque and 30 N of force. To allow the torque caused by the spinning of the friction roller to be modeled, we simplified the CAD model to include the friction roller and motor as one component, with the other being the housing. Under maximized loading scenarios, the stress and deformations are shown in Figures D6, D7, and D8.

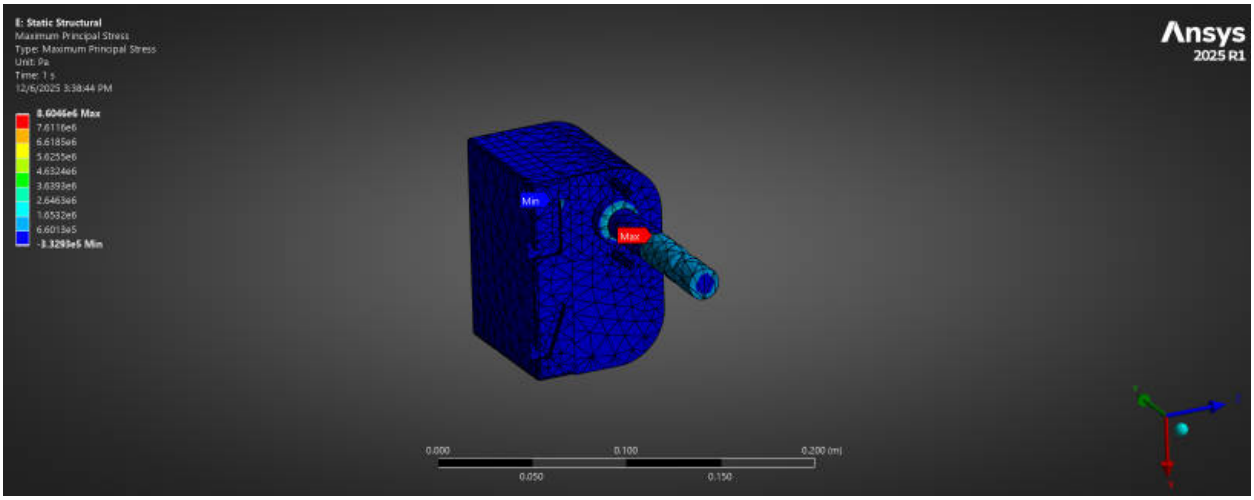


Figure D6: Max Stress on Friction Roller

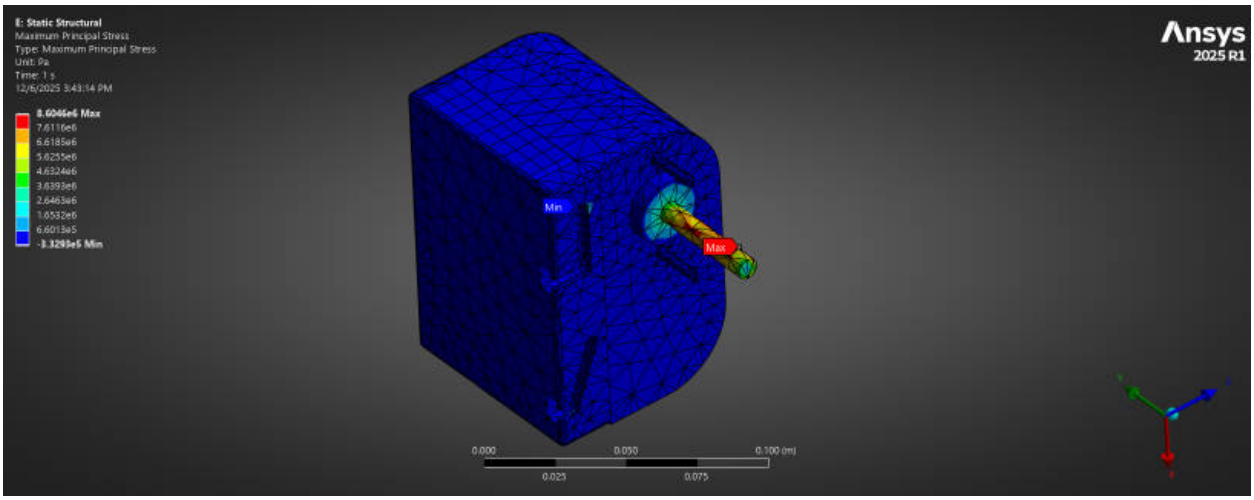


Figure D7: Max Stress on Motor Shaft:

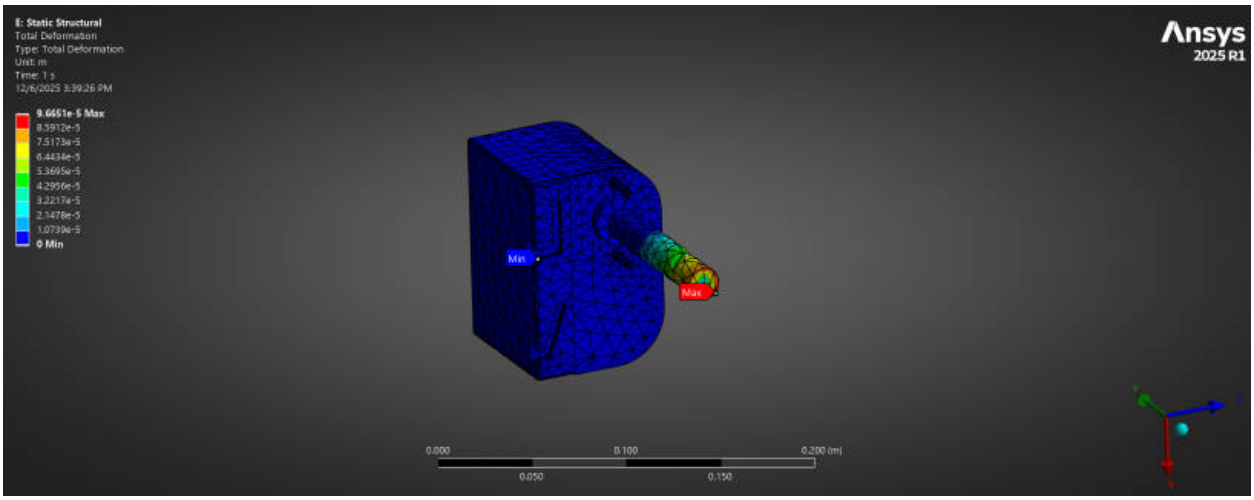


Figure D8: Total Deformation:

The maximum yield stress for PLA is 30-60 MPa. Based on these results, the device is safe to use and will not fail due to stress. Based on the model, the max stress on the PLA is 1MPa, far below the maximum. For the motor shaft (steel), the maximum stress reaches 8.6MPa, which is far less than the maximum yield stress of 2GPa. Based on this analysis, the H.P.C.B. will not fail due the torque caused by the spinning of the friction roller.

Appendix E: Model Based Production Planning

Our company starts with an initial production of 75 units with cost of materials adding up to $50 per unit. We plan to list our product at $80 providing payment terms of 30, 60, and 90 days before payment is due. A fixed overhead of $15000 for business expenses have been set aside with a downpayment that is 50% of the revenue per week. For our model based production planning, our initial release of the product came with preparations for production to exceed demand. We expect stable operations throughout our first year as we steadily gradually increase production to 15 units per week/ ramp up rate. Looking at the data in figure E2, we expect a demand increase by 4% per week where production is able to keep up with demand until week 85. Our product is in its first model and as we reach high demand near the closing weeks, we expect people to be interested but cautious about purchasing.

As our company provides 3 different payment terms, data of our cashflow is charted as all payment terms are starting at an initial net cashflow of -$15500 from overhead, downpayment, and cost of goods sold. In figure E1, the trajectory of net cashflow will be similar throughout all payment terms with 30 days providing a headstart. Our company expects to be in the negative for over a year as we continue to operate expecting the shift around week 60.

Our manufacturing process heavily relies on 3D printing as our product's frame is designed using PLA filament. With our equipment currently, 15 units per week can be easily done and as stated before, production will exceed demand for the majority of the 100 weeks.

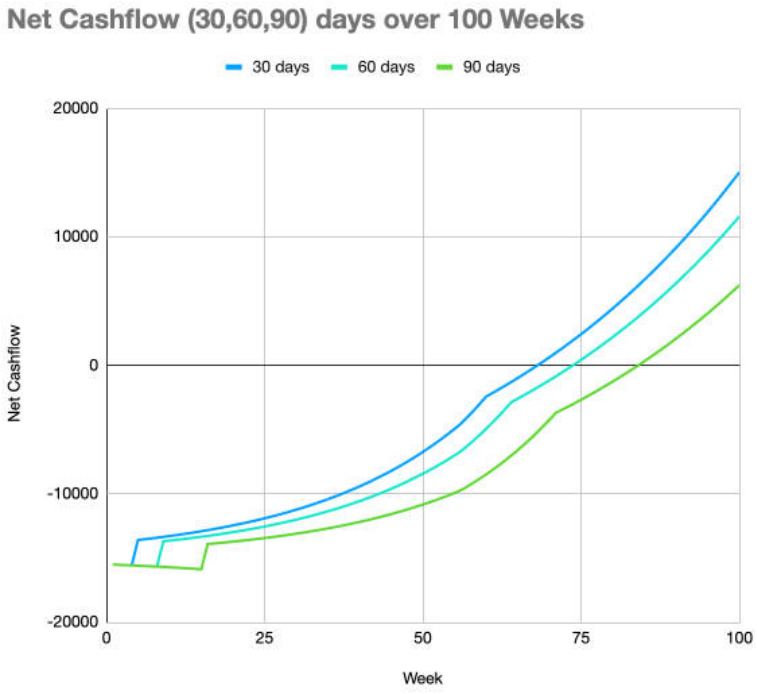


Figure E1: Net Cashflow including payment terms 30, 60, 90 days over 100 Weeks

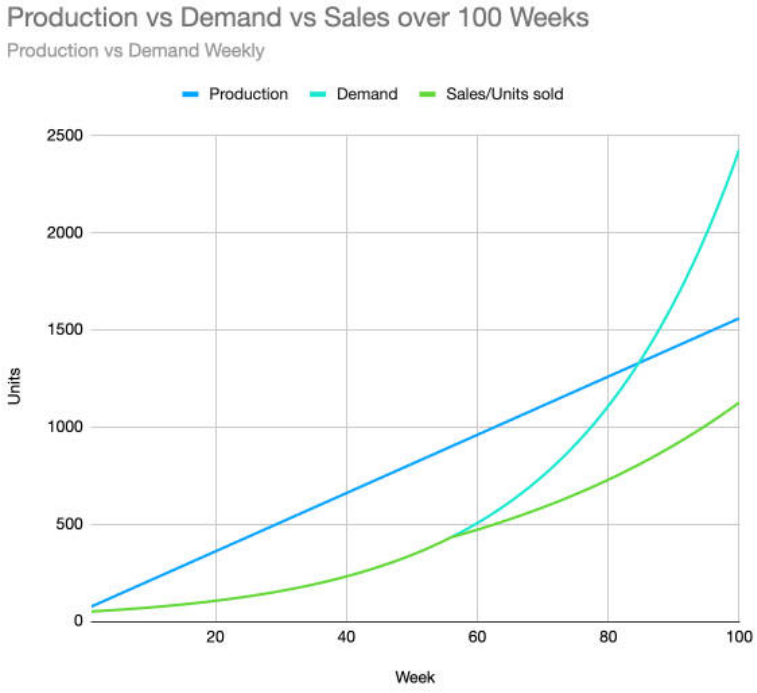


Figure E2: Production vs Demand vs Sales of Units over 100 Weeks

Appendix F: Virtual Ecosystem

Our project values sustainability to the highest standards as it is our responsibility to ensure our model upholds it.

Our model has performed as calculated generating mechanical energy that is converted to electrical power from the H.P.C.B. as the bike is in rotational motion. The H.P.C.B. device's self-sustaining power generation removes the requirement of replacing batteries. Environmental and Sustainability of our project minimizes waste using a stationary portable power bank. Reusing a powerbank reduces our carbon footprint by reducing greenhouse gas emissions. Gathering materials such as lithium for batteries is not environmentally friendly and should therefore be limited as much as possible while using long lasting materials. This also lowers our carbon footprint reducing the manufacturing production time and material extraction tools usage. Recharging batteries to power bike lights reduces disposable battery usage and ensures human safety.

Our Virtual Ecosystem in Figure F1 showcases many beneficial environmental impacts on land, water, and air. The H.P.C.B. device reduces E-waste since the user is riding on a bike there will be no emissions. The virtual ecosystem showcases end of lifecycle benefits for the H.P.C.B. device and bike as local bikes shops will repair damaged bikes while unrestorable bikes can be recycled. The H.P.C.B. model can easily be disassembled and entirely user friendly to be replicated as we provided the 3D printing model of the casing. The PLA filament is recyclable. Cables and electronics can be replaced on third party markets like Amazon.

Users and bike shops may provide valuable feedback to improve future designs and continue to work on sustainability delivering feedback straight to the manufacturers.

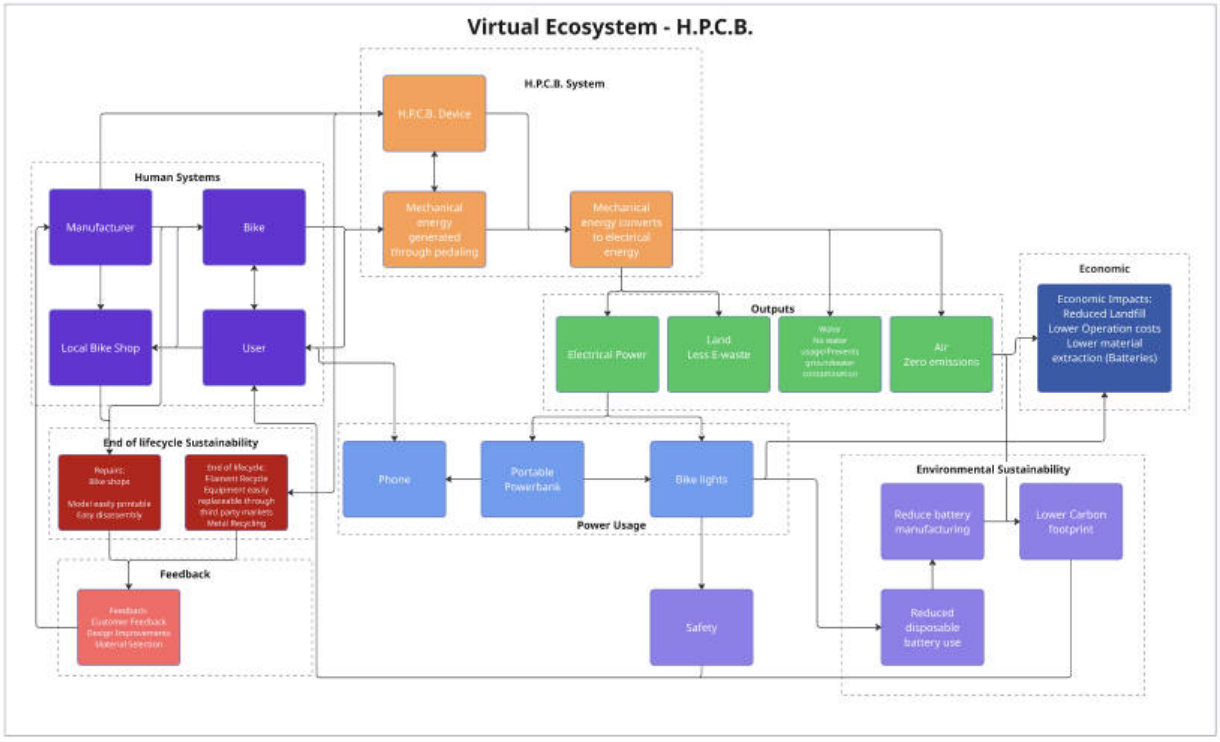


Figure F1: Virtual Ecosystem Block Diagram

Appendix G: As Needed and As Offered

Problem Statement: To convert a portion of energy from cycling into useful electrical energy for charging devices in an unobtrusive way.			
#	Ideal (As Needed)	Revised (As Offered)	Justification
1.	The system shall generate enough energy to at least maintain a battery percentage on a modern smartphone while running GPS.	While biking at 10 mph, the device shall generate sufficient energy to charge a modern smartphone.	Made more specific and removed the "GPS" requirement, as energy required is highly variable on phone, app, and settings.
2.	The system shall be light enough to not be obtrusive to bikers.	The system shall be weigh no more than 1 kg (approximate weight of a full cycling waterbottle)	Made more specific so that the result could be measured.
3.	The system shall not noticeably slow the bike and biker down.	The rear wheel, with the device attached, shall remain able to be spun freely at a minimal torque applied (~1 Nm) when the wheel is suspended in the air.	Made more specific so that the result could be measured.
4.	The system shall be water resistant, at least enough to be used in rainy conditions.	The device shall be positioned such that it is shielded from rain and splatter from the wheel, it shall have all joints waterproofed via sealant or gaskets.	Made more specific so that the result could be measured.
5.	The system shall absorb vibrations, so as to not become damaged in bumpy conditions.	The device shall absorb or withstand typical vibrations experienced while cycling, and shall be able to withstand riding over bumpy terrain such as gravel.	Made more specific so that the result could be measured, added withstanding vibrations as it is reasonable for the amount of high frequency vibrations that occur when cycling.
6.	The system shall be able to be mounted on the 90th percentile of bicycles via adjustable mounts or a simple universal mount.	The device shall have universal mounting options for all standard adult bikes (27.5 in wheels), excluding full suspension frames.	Made more reasonably specific so that the result could be measured, excluded common non-standard frames as it lacks water bottle mounting points (otherwise universal).
7.	The system shall be almost entirely encased to provide protection while also providing a sleek design.	The system shall be almost entirely encased to provide protection while also providing a sleek design.	Unchanged
8.	The system shall contain an internal battery capable of providing a charge even when the bike is stopped.	The system shall contain an internal battery capable of providing a charge even when the bike is stopped.	Unchanged

9.	The system shall include visual indicators to let the user know how much battery is stored at any given time.	The system shall include visual indicators to let the user know whether the battery is charging.	Removed ability to read battery level, as it is outside the scope of a 10 week design cycle and would require fabrication of custom battery pack (complex battery chemistry and internal regulators prevent simple readings)
10.	The system shall include at least 2 USB-C power output ports.	The system shall include at least 1 USB-C power output port.	Reduced the requirement to 1 USB-C power output port, due to the price of battery packs with additional power ports.
11.	The system shall be durable enough to be used on the bike for at least 5000 miles.	The system shall be able to withstand tests on pavement, dirt, gravel, and mounting/dismounting a curb, without showing any signs of wear.	Changed durability requirement so that it could be tested within the time period of 10 weeks.
12.	The system shall not exceed the cost of $200 to manufacture.	The system shall not exceed the cost of $50 to manufacture.	Reduced budget requirement to be nearer to the true cost of parts, while leaving some leeway (excluded parts purchased but unused).
13.	The system shall not require any specialized modification to the bike, and should be able to be installed by anyone at their home with common household tools.	The system shall not require any specialized modification to the bike, and should be able to be installed by anyone at their home with common household tools.	Unchanged
14.	The system shall be able to regulate its temperature via proper ventilation, and should automatically shut off if it becomes too hot so as not to be a danger.	The system shall have a sleep mode, so that it powers off automatically when the bike is not used. It shall have the ability for the sleep mode to be enabled/adjusted/disabled.	Altered requirement to include sleep mode for power efficiency rather than temperature safety, since we discovered the temperature was not a real concern within our system.
15.	The system shall be able to function whether it is in hot or cold temperatures, without significant fluctuations to its performance.	The system shall be able to function within temperatures ranging from -5 ℃ to 30 ℃.	Made the requirement more specific, containing a reasonable range of temperatures bike riders could experience (on the extreme side).
16.	The system shall include both a wheel generation system and a regenerative braking system, the latter of which should be able to recover at least 10% of kinetic energy dissipated in braking.	The system shall have a removable battery, such that it could be used as a regular battery pack while the H.P.C.B. is not in use.	Changed requirement to not include regenerative braking, which made requirement redundant with requirement 1. Added a new requirement for battery removability.

17.	The system shall attach to the bike in a secure enough manner where it is not easy to steal while the bike is stored outside.	The system shall attach to the bike such that it does not fall off while being ridden, and such that it cannot be removed without tools/expertise.	Changed requirement such that it does not contradict requirement 13.

Table G1: As Needed and As Offered Requirements

The ideal project requirements were preserved for the most part, and were primarily adapted so that they would be more testable with our final design and short design cycle. One requirement was removed (16), and replaced with a different one, as it was outside the scope of what we could design within the 10 week period.

Appendix H: Pre-Mortem

Due to the composition of our product consisting mainly of PLA, the manufacturing process is the largest area in which error could occur. Despite our best efforts to design each hole for maximum printing precision, due to the nature of 3D printing, sometimes perfect tolerancing is just not possible. The larger and more dependent the case is on 3D printing, the higher the likelihood that printing errors can arise. Because we only had to create a single product for this class, any errors can be easily taken care of with time but if we were to be on the scale of producing 100 shells a day for example, then we would need to flesh out the manufacturing process more. The most concerning part for the manufacturing process would be all the wire soldering and attaching the various components to the case itself. Once again, because we only had to make one product and were given ample time to do so, any small errors that were made such as a poor solder or not enough glue to hold the wires down were easy to fix. On a large scale production, we would most likely need to redesign the base to be able to either snugly fit each separate component or be able to have specific instructions on how much glue to apply and to what parts to ensure a snug connection along with ample vibration absorption which is a critical need for our product to work. As for what could go wrong for the product itself, the main factors of concern would be the casing or wiring. Because bike rides can be extremely rough, it is possible that longer and bumpier rides could loosen bolts over time and cause the entire H.P.C.B. to fall off the bike mid ride. If not falling completely off, then it could offset the tire to tire connection making the efficiently rapidly drop or even stop producing electricity all together. This is another reason why the vibration absorption is so important so if the absorption methods implemented get damaged or misplaced, it could threaten the entire system. A fall off the bike or large amount of pressure could also crack the case which would also compromise the entire

system. If any water, dirt, or debris were to enter a crack and bypass the waterproofing of the design then many of the electrical components would cease to function breaking the entire electrical circuit. Because our current electrical circuit depends on each part to function properly, any damaged parts would compromise the system even if the user didn't care for the part. The final thing that could go wrong with our product design would be user based. If the user fails to properly attach the H.P.C.B. to the bike then that would be catastrophic to integrity. The user could also fail to properly align the two tires which would make efficiency drop drastically. A final example of user error would be forgetting to shut the system off which might end up draining the battery pack by continuing to draw power from it to power the ESP32 despite it not being needed. Overall, while our product does have multiple post-mortem concerns, as long as we are aware of them and design around them, our product will have high reproducibility rate and user success.

Appendix I: As Built



Figure I1: H.P.C.B. Complete Prototype on a size XXL Road Bike



Figure I2: H.P.C.B. Complete Prototype on a size XXL Road Bike Close-Up



Figure I3: Complete H.P.C.B. Prototype Including Optional Zip-Tie Mounts



Figure I3: Complete H.P.C.B. Prototype with Friction Joints Exercised

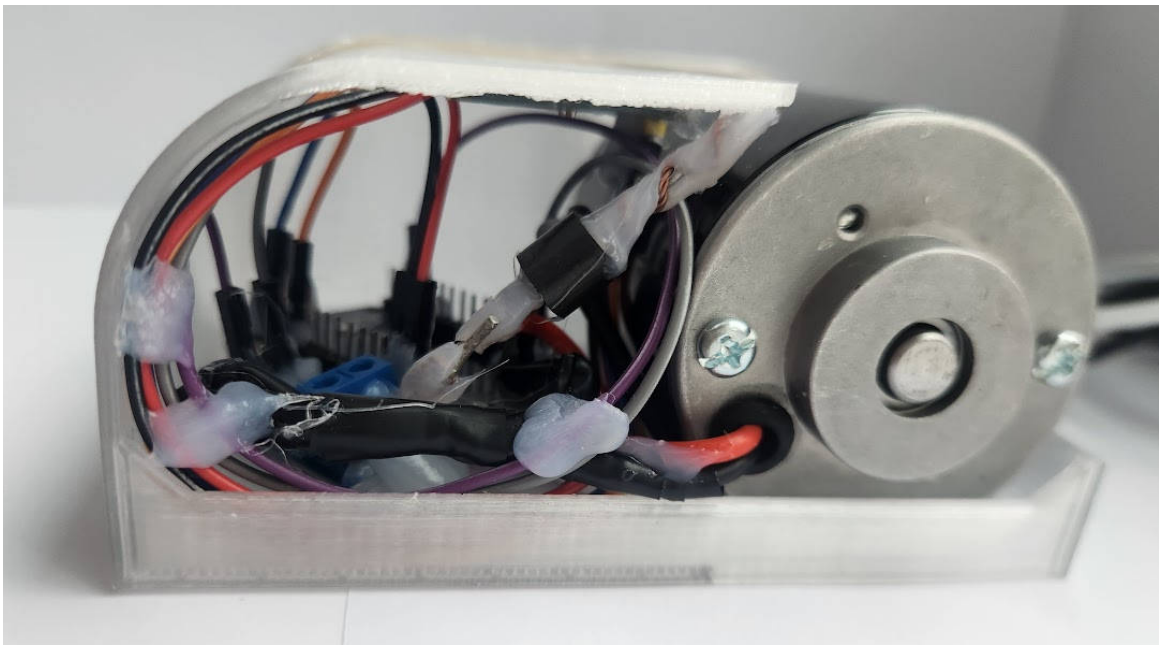


Figure I4: H.P.C.B. Removable Insert Containing Electrical Components



Figure I5: H.P.C.B. Removable Insert Showing Rubber Gasket

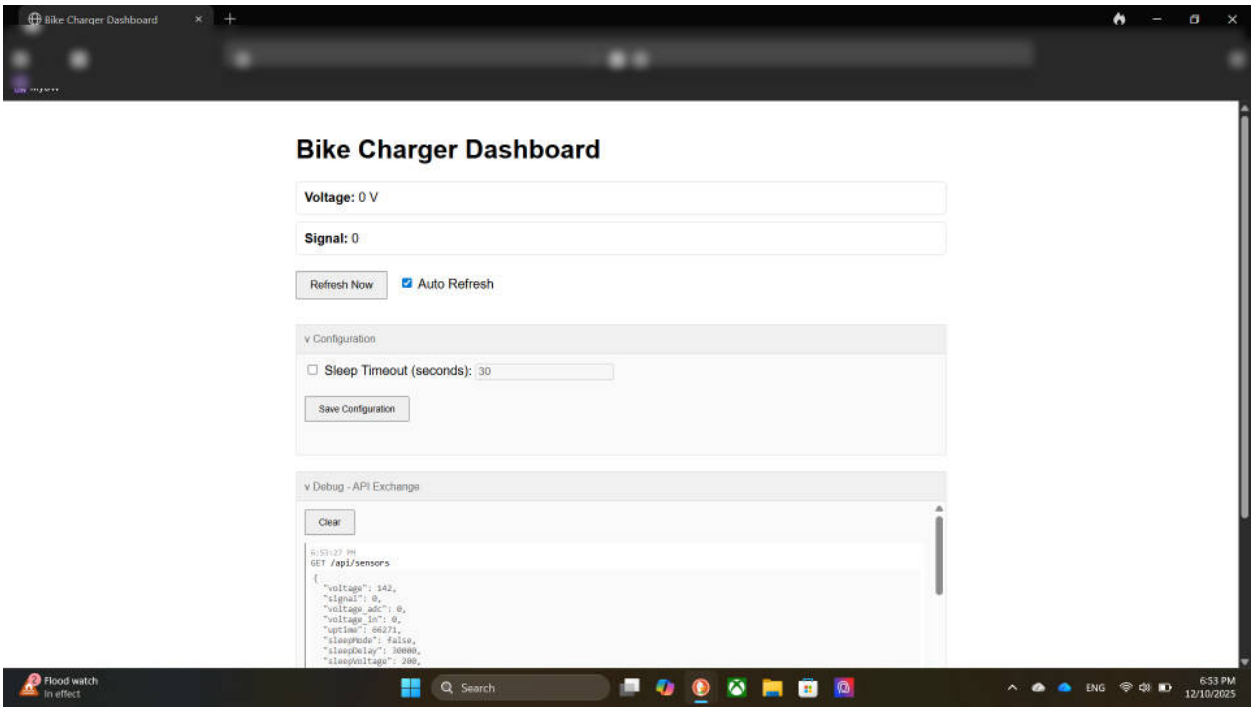


Figure I6: H.P.C.B. Website Interface



Figure I7: H.P.C.B. Display With Sleep Mode Enabled

Appendix J: As Tested

Wattage Output Test:

A voltage output test was performed to see whether the H.P.C.B. is capable of outputting a useful amount of wattage under normal riding conditions. For these tests, a constant current of 0.5 Amps was assumed, as that is the current we measured that the battery begins charging at. We found a linear relationship between bike speed and wattage output, which lines up with our calculated equation in Appendix A, Figure A1. This relationship is shown in Figure J1.

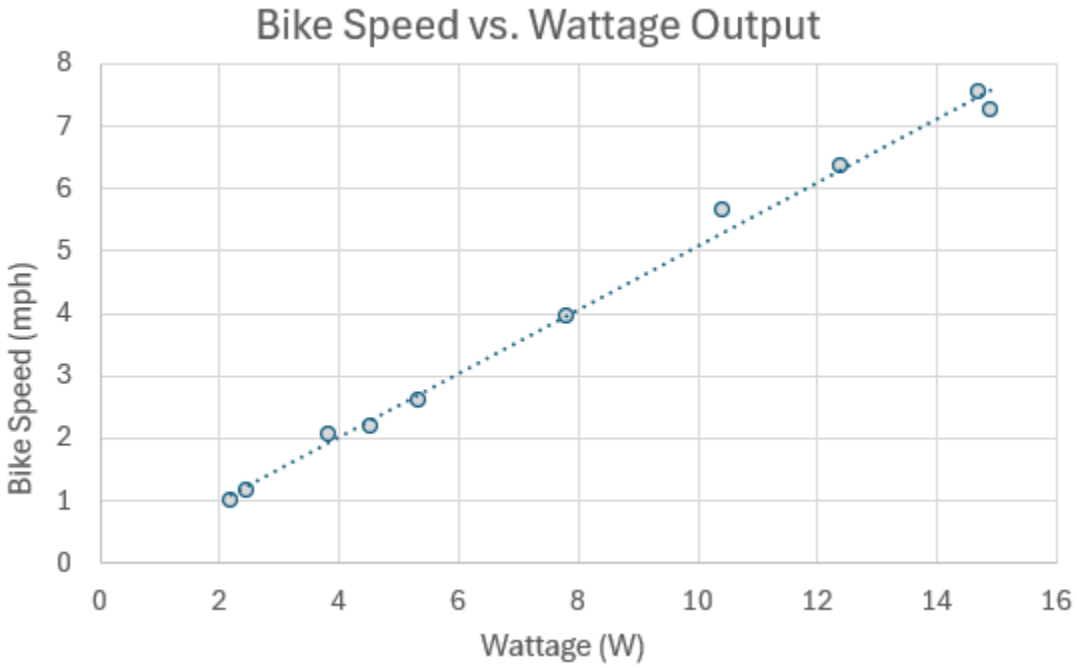


Figure J1: Bike Speed and Wattage Output Relationship Graphed

This graph was obtained from values measured on flat ground over the course of 10 trials. The table of these values is shown in Table J1.

	A	B	C	D	E	F	G	H	I	J	K
1	**Voltage Output Test**										
2											
3		**Trial**									
4		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
5	**Speed (mph)**	10.4	4.52	3.81	14.9	2.19	12.4	14.7	2.46	5.32	7.82
6	**Voltage Output (V)**	11.3	4.36	4.12	14.5	2.02	12.7	15.1	2.32	5.21	7.9
7	**Current Drawn (A)**	0.5									
8	**Wattage Generated (W)**	5.65	2.18	2.06	7.25	1.01	6.35	7.55	1.16	2.605	3.95
9											
10	*Test performed at various speeds due to difficulty noting at which speed a particular voltage was achieved, especially while operating a										
11	bicycle.										
12	*Speed measured via mobile fitness application.										
13	*Uncertainty in measurements caused by delay between recording speed voltage										

Table J1: Voltage Output Test Results

An image showing a test being performed is shown in the figure below; J2.



Figure J2: Stillframe of Voltage Output Test, Performed in Real-World Scenario

Durability Tests:

A series of durability tests was also performed. In these tests, the device was mounted on a size XL mountain bike and ridden over various terrain for a duration of time around 10 minutes. After the ride, qualitative observations were recorded for the state of the H.P.C.B. device, noting any potential damage or weak points. These tests were performed on pavement, dirt, and gravel

surfaces. The results of these tests are shown in the Tables below. A figure of each test location is also included below.

	A	B	C
1	**Pavement Test**		
2			
3	**Component**	**Qualitative Observations**	**Time Spent Riding**
4	Housing	No significant effects. Remained fixed in place.	10 min 13 seconds
5	Insert	No significant effects. Remained fixed in place.	
6	Friction Roller	No slipping, excellent hold onto tire.	

Table J2: Pavement Durability Test Results



Figure J3: Pavement Durability Test Location

	A	B	C
1	**Dirt Test**		
2			
3	**Component**	**Qualitative Observations**	**Time Spent Riding**
4	Housing	Experience significant vibrations, did not slip along mounting rails.	10 min 13 seconds
5	Insert	Remained fixed in place, no sign of slipping.	
6	Friction Roller	Accumulated dirt on tire, no visible slipping	

Table J3: Dirt Durability Test Results



Figure J4: Dirt Durability Test Location

	A	B	C
1	**Gravel Test**		
2			
3	**Component**	**Qualitative Observations**	**Time Spent Riding**
4	Housing	Shook significantly while riding, but not slippage along mounting slots.	10 min 5 seconds
5	Insert	Remained fixed in place, no sign of begging to fall out.	
6	Friction Roller	Accumulated a significant amount of dust on the tire, rare slipping with tire.	

Table J4: Gravel Durability Test Results



Figure J5: Gravel Durability Test Location

An additional durability test was performed with the bike being ridden up and down a curb, as that is a common scenario for cyclists. The curb was ridden up and down a total of 10 times, qualitative observations were recorded after all 10 curb mounts/dismounts were completed.

These results are shown in Table J5, and a picture of the curb is shown in Figure J6. In Figure J6, the tallest part of the curb is where the mounting and dismounting durability was tested.

	A	B	C	D
1	**Curb Test**			
2				
3	**Component**	**Qualitative Observations**	**Times Down Curb**	**Times Up Curb**
4	Housing	Shifted up and down along mounting slots, could likely be fixed by tightening mounting screws.	5	5
5	Insert	Remained fixed in place, did not being to slide out or show any signs movement.		
6	Friction Roller	Tires remained fixed in their position on the axel. While mounting and dismounting, connection between tire and wheel occasionally slipped, but was repositioned after landing.		

Table J5: Curb Durability Test Results



Figure J6: Curb Durability Test Location

Based on these tests, the H.P.C.B. device is capable of providing a useful amount of wattage at reasonable bike speeds. Additionally, it is durable enough to withstand common riding surfaces and conditions.

Appendix K: As Certified

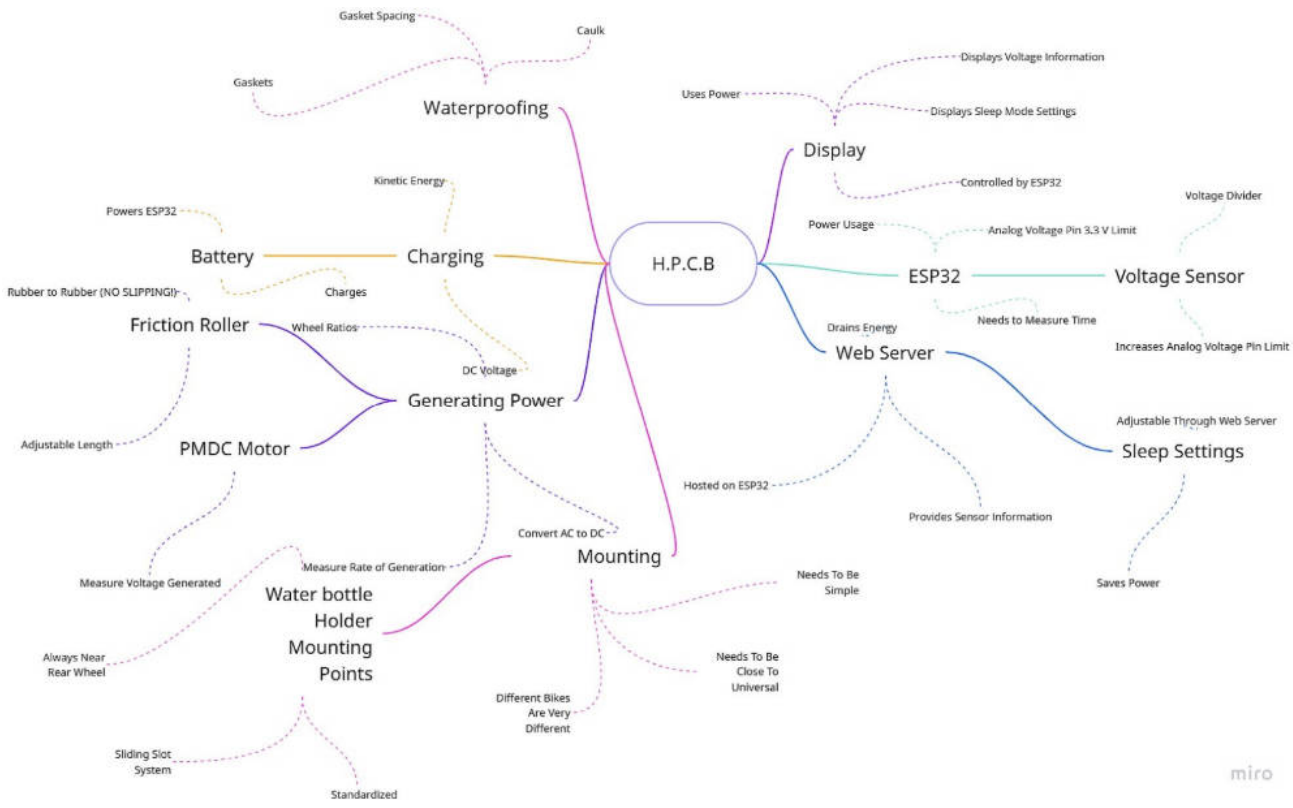


Figure K1: Mind Map

Looking at Figure K1, our mind map, you can see that we had a few main aspirations for the project and then from there built off of them one by one. For example, we aspired to make our final product completely waterproof but only later did we decide how that would be done. By basing our design off our mind map, the possibilities of re-design and material changes are endless. Because each individual branch of the map represents its own section, unaffected by any other section, it is extremely easy to make modifications and improvements to our current design. One possible design change that could be made is to add a phone holder to the bike's handle bars. While our current design boasts an extremely portable frame, it only holds the battery pack itself. It is still completely up to the user to stop riding, take the battery pack out, and to charge their phone if it dies at any location they stop at to charge. If we simply add an

attachable frame and possibly a wire holder that would connect the wire to the H.P.C.B.and up to the phone, it would greatly increase the user experience. The user would be able to ride without stopping and have access to, say, google maps while riding and any other notifications. This also brings in the possibility of adding to the holder a screen of the volts they produce or the mph of the rider based on the wheel velocity. Because of the centralized mind map design, additions like this are easy to add on. As for material changes, this is where the quality of the product can greatly improve. Our current design is based on two things: portability and cheapness. If we simply want to upgrade our materials from PLA to a higher grade plastic or polymer, then the sturdiness and toughness of the case will increase while maintaining its light weight. One other material change that could greatly improve efficiency is the rubber tire connector to the main back tire. Because we are currently using low grade rubber (the ones used to make lego tires), they degrade and wear with only a few hours of usage. By switching out to a high grade rubber or epoxy, the tire to tire traction could improve and along with that the effectiveness of the connection would improve along with charging speed and power generated. The next steps that we have for improving the design would be multiple iterations of possible additions and material changes as the examples just stated. Due to the simplicity of the design, there are not many necessary changes needed to be done to get a more complete product. The product works as intended and does a good job at it. When looking at any material changes or design changes, it becomes more a conversation of what should be done in addition to our already existing product. Because material changes often increase cost, the addition of new parts is the best solution as that would increase revenue and attention given to the main product.

Appendix L: As Delivered

Upon arrival, the product inside is secured to a custom made recyclable cardboard insert ready to be unboxed and installed onto any bike. The product is in pristine condition with a freshly printed PLA filament casing in the color of the buyer's choosing. The H.C.P.B. can be seen in figure L2 of the ideal model. Accessories included inside the box are a mounting bracket, cables, USB-C, and instructions on how to install onto a bike. The initial setup of the H.C.P.B. is to install the mounting bracket next to the back wheel on the frame under the seat. The H.C.P.B. shall be inserted into the mounting bracket with the friction rollers against the bike wheel. Connect the cable to the battery. The user begins to pedal and the H.C.P.B. will automatically turn on as voltage is generated and the percent of battery charged is displayed on the screen. In figure L3, the BOM table provided consists of the parts used to create the H.C.P.B.

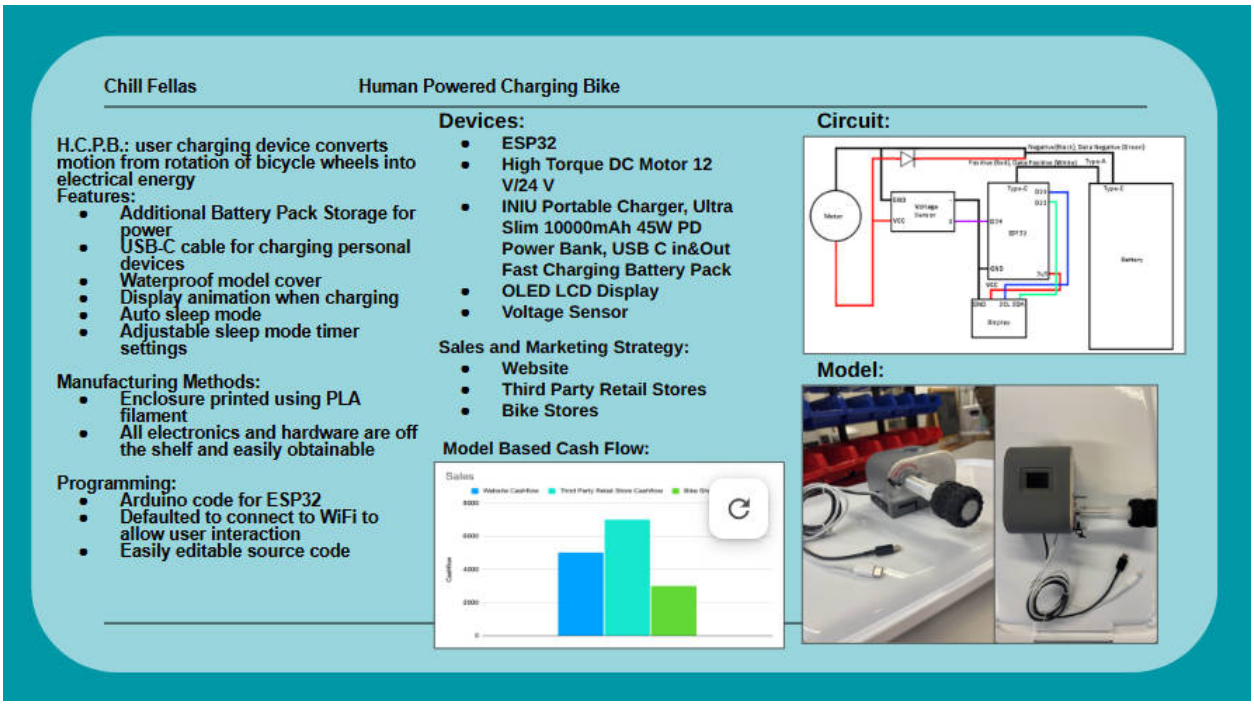


Figure L1: H.P.C.B. Walkaround Chart



Figure L2: Render of Ideal H.C.P.B.

# Item Number	Part Name	Manufacturer	Qty	Description
1	DC Motor	Honses	1	High Torque DC Motor 12 V/24 V
2	Voltage Sensor	DORHEA	1	Monitors voltage being generated
3	USB-C Cable	MEOU	2	Connects battery pack to electronics
4	Friction Roller		1	Attached against bike wheel
5	Housing	Chill Fellas	1	Waterproof casing surround system
6	Insert	Chill Fellas	1	Waterproof casing surrounding electronics
6	Rectifier Diode	OTOTEC	4	allows current oneway
7	Battery Pack	Shenzhen Topstar Industry Co.,Ltd.	1	Stores generated energy
8	M4 Nut		2	
10	Display Screen		1	Waterproof screen protects OLED Dispaly
11	OLED Display	HiLetgo	1	Displays Voltage
12	ESP 32	ESPRESSIF SYSTEMS	1	Brain of the whole assembly using Arduino programming
13	M3.5 14 mm Screw		1	
14	Rubber Band		3	

Figure L3: H.P.C.B. BOM Table

Appendix M: As Supported

The H.P.C.B. device that we have created was built for reliable use; a product that could survive in any terrain, whether it be a gravel path, dusty road, or even pouring rain. Each hole and opening was either waterproofed with sealant or is airtight with rubber fittings. The rubber at each connecting point to the bike is also for vibration absorption. Any loose wire has been connected firmly through soldering and then after soldering is wrapped in rubber to ensure no loose wiring. The most likely point of failure would be during a crash. If you are mountain biking or you simply fall hard and land on the H.P.C.B., the PLA that it is made of would most likely crack. Any cracks in the casing would lead to multiple other problems such as making the H.P.C.B. susceptible to water, dirt, and debris. It would also make the H.P.C.B. no longer vibration resistant, and any other further vibrations would only serve to worsen any damage sustained to the case. In the situation of the case cracking, the H.P.C.B. would need to be taken in for repair. Another possible point of failure, failure in this case being a large drop in power, would be the rubber tire connectors wearing. If used for a long period of time, the rubber connecting tires we use to connect to the bike main tires showed signs of wear. This might be due to the cheap material used, and would be a point of improvement if we were to take this design further. Both of these failure points however would have extremely simple fixes. Because a large majority of the case is 3D printable, if your case cracks then all you would have to do is 3D print yourself a new case. If you do not have access to a 3D printer then you could also simply purchase a new case from an in person store assuming there would be a franchise. The only case of servicing that would need to be completed would most likely be if any of the electrical components were damaged. What would need to be done is a simple diagnostic to test each component performed by an employee and once a part is found to be faulty, that part would

simply be removed and replaced for a working part. Maintenance of the H.P.C.B. would be the simplest part. Because it is virtually a single case with no access to the delicate insides, all one would need to do is wipe the casing. Maintenance would virtually be dismissible for our product. The only point of optional maintenance you could do would be to replace the little rubber tires after you see significant wear on them. In this case you would simply unscrew the U.P.C.B., remove it from the bike, slide out the rubber tires, and replace them with brand new ones that you can easily purchase for cheap from a main store. The final point to cover would be how the customer would reach the aforementioned points of maintenance and service. The easiest form of this would be to actually bring in the H.P.C.B., to a physical store to have a professional take a look in person and perform a diagnostic there. Due to the relative simplicity of the design, these diagnostics would not take a long time and would be few and far between for the customer. Another point of contact would be our main website and company phone. If the problem is user error for example, one simple phone call would do the job. You could also use the website as a guide point to either make appointments, find stores, and also guides on maintenance and correct usage for the product.